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**SIGNALITE AND HDe PATIENT TRANSPORT**  
**INSTALLATION MANUAL**

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|                                                                                                                                                                                                   |                         |                                                                                                         |            |           |              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------|------------|-----------|--------------|
| Approved and<br>Reviewed by: <b>Anand K</b>                                                                                                                                                       | Date: 27/Oct/06         | <b><u>SIGNALITE AND HDe PATIENT</u></b><br><b><u>TRANSPORT</u></b><br><b><u>INSTALLATION MANUAL</u></b> |            |           |              |
| Author: <b>Kiran Kumar M R</b>                                                                                                                                                                    | Date: 04/May/09 (Rev 5) |                                                                                                         |            |           |              |
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|                                                                                                                                                                                                   |                         |                                                                                                         |            |           | Rev 5        |

| Rev | Date        | Author      | Change  | Description                                                                              |
|-----|-------------|-------------|---------|------------------------------------------------------------------------------------------|
| 1   | 23/01/2004  | Narendar.V  | -       | Rev 1 release                                                                            |
| 2   | 07/03/2005  | Narendar. V | -       | Up sensor pics updated.                                                                  |
| 3   | 27/Oct/2006 | Kiran       | Revised | HDe Table Added and sections updated                                                     |
| 4   | 04/Jan/2008 | Kiran       | Revised | Section 17-Front guide rollers adjustment added and Section 6 updated                    |
| 5   | 04/May/09   | Kiran       | Revised | Section 3.3-Checking Caster Height Adjustment Modified for height adjustment dimensions. |

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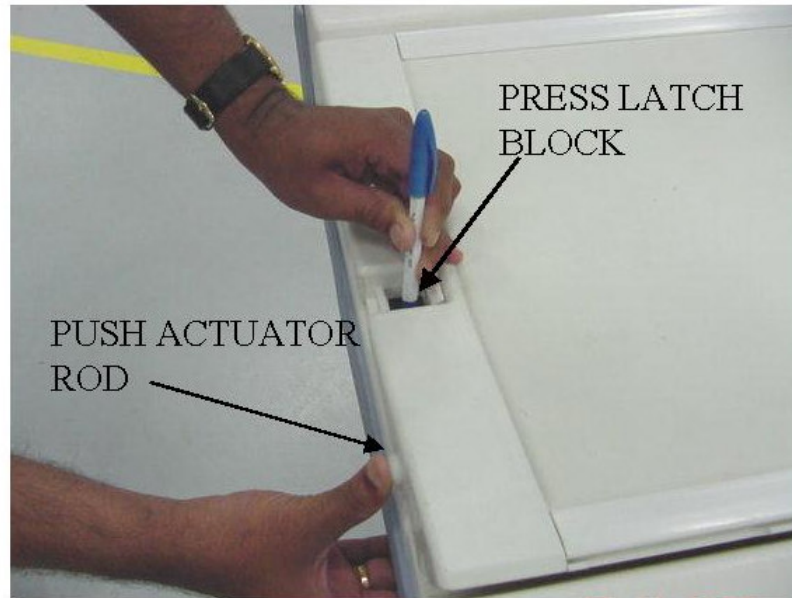
**Note:** This Section contains all adjustment procedures, which may need to be performed on Signa lite patient transport; however, most of the adjustments are factory preset and should not require adjusting in the field. The adjustments should only be performed:

- As part of installation as specified by the installation procedures,
- After determination of a problem via the functional check procedures in this Manual.
- After a component replacement as specified in the replacement procedures.

During the installation procedure of the patient table there may be a requirement to pull up of the bellows to service position. Refer to Section 2-1 pulling up of bellows to service position, on Patient Table; some replacement procedures also require the removal of the cradle. Refer to Section 1 below - for cradle assembly removal.

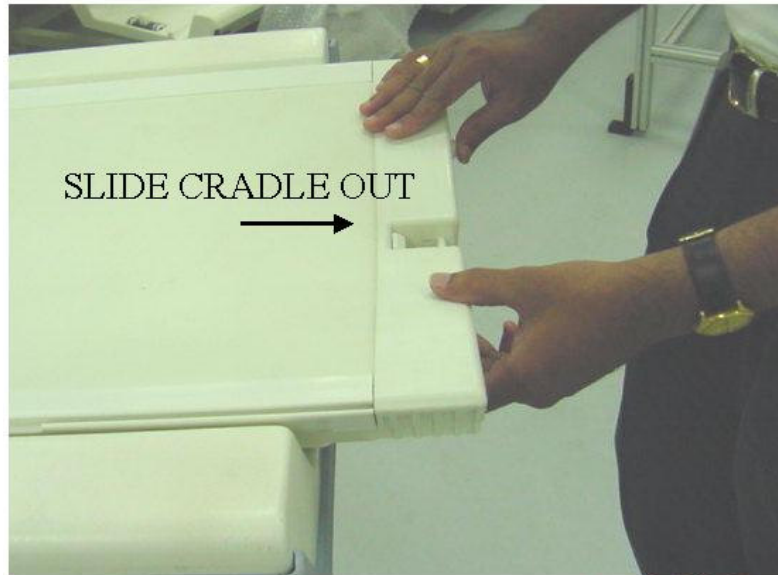
## **1. REMOVING CRADLE ASSEMBLY**

- 1.1.** Depress cradle latch block at front end of table with a blunt object such as a pencil eraser or pen end, and push in actuator rod at front of cradle, at the same time. Refer Illustration 1.



**PRESSING ACTUATOR ROD AND LATCH BLOCK  
ILLUSTRATION 1**

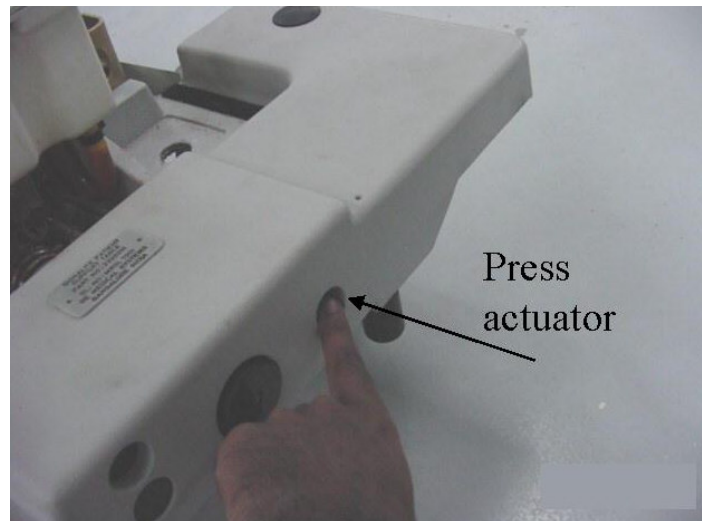
- 1.2. While holding these actuators, slide cradle forward on patient table approximately three inches. The actuator can now be released. See Illustration 2.



**SLIDING CRADLE OUT THREE INCHES**

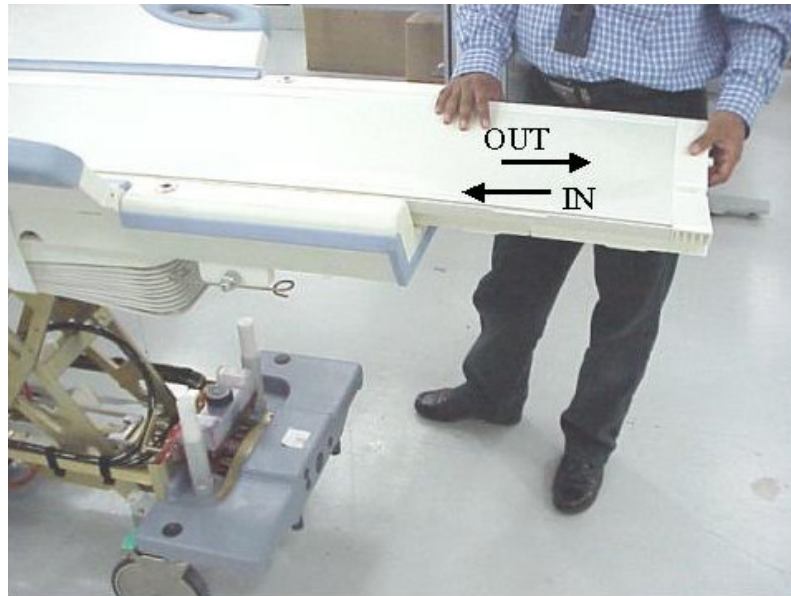
**ILLUSTRATION 2**

- 1.3. Push the actuator located on right side of Front casting and slide cradle completely out of Table Top. Keep Cradle aside safely. See Illustration 3 and Illustration 4.



**PUSHING ACTUATOR AT BASE**

**ILLUSTRATION 3**



**PATIENT TABLE CRADLE REMOVAL  
ILLUSTRATION 4**

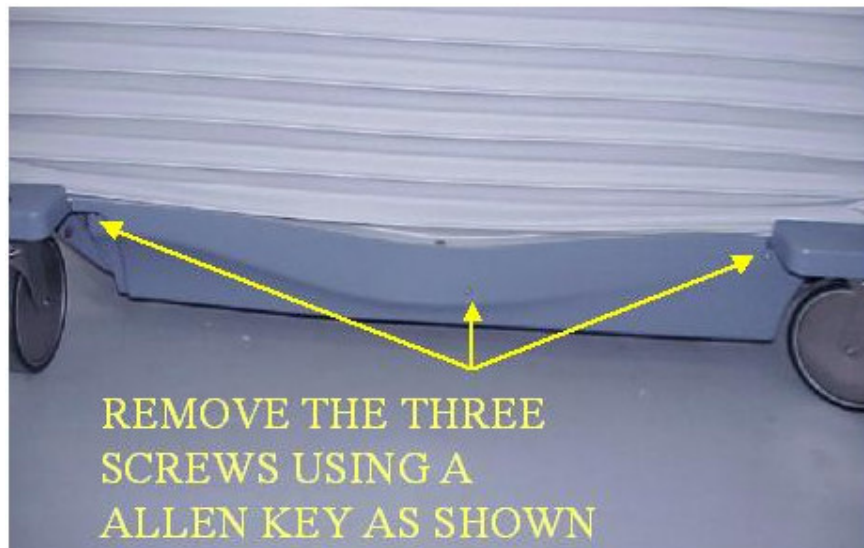


**Note:** Instructions given below apply from Table Serial No. 2822WH5 onwards.  
For Tables prior to the specified Serial No, Please refer to annexure, attached at the end of this manual.

## 2. BELLOWS

### 2.1. PULLING UP OF BELLOWS TO SERVICE POSITION

- 2.1.1. Undock patient table, and move it out of exam room.
- 2.1.2. Remove the three screws of the bottom cover using an Allen key as shown (Ref illustration 5A below). Use the same procedure to remove the bottom cover on the other side of the table also.



**BOTTOM COVER REMOVAL**  
**ILLUSTRATION 5A**

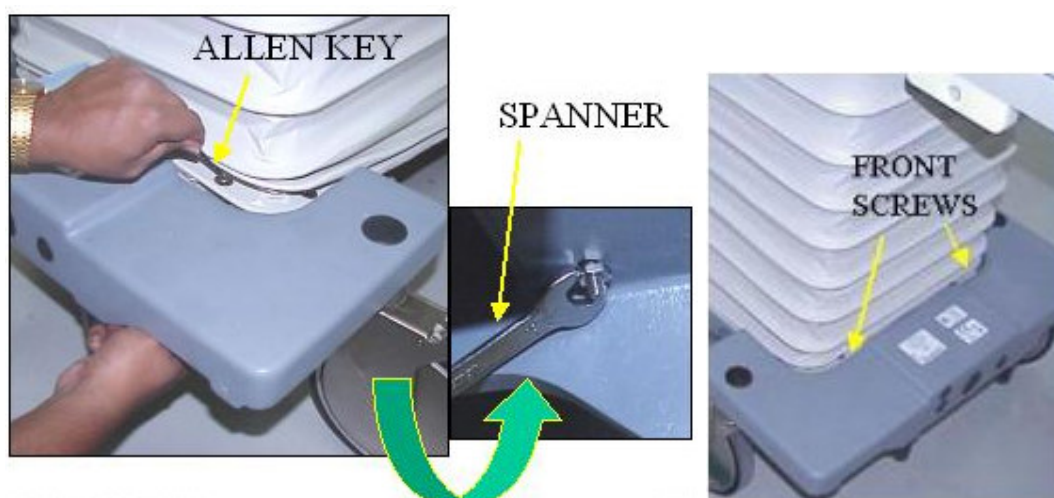
- 2.1.3.** After removal of screws, hold the bottom cover and remove it as Shown. Refer to Illustration 5B. Remove the bottom cover on the other side of the table also, using the same procedure.



**REMOVAL OF BOTTOM COVER**

**ILLUSTRATION 5B**

- 2.1.4.** Remove the two screws fixing the bellows on the front casting. Refer to illustration 5C. To remove these screws use an Allen key to hold the screw and loosen nut below using spanner. Refer to illustration 5C. After removal of the screws leave the bellows support strip in place for reassembly later. Refer illustration 5D.



**ILLUSTRATION 5C**



ILLUSTRATION 5D

- 2.1.5.** The bellows are fixed on three brackets on either side, using a screw and nut. Refer illustration 5E. Using an Allen key hold the screw and loosen the nut using a spanner. Refer illustration 5F. Repeat the procedure and remove the screws on the other side of the table also.
- 2.1.6.** The rear end of the bellows is fixed onto the rear casting using Velcro. Remove the bellows at this end by lifting upwards as shown. Refer illustration 5G.

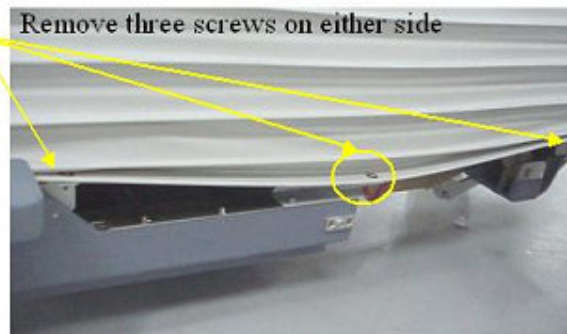


ILLUSTRATION 5E

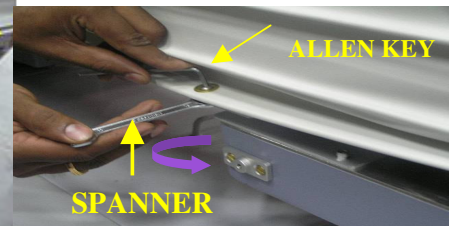


ILLUSTRATION 5F

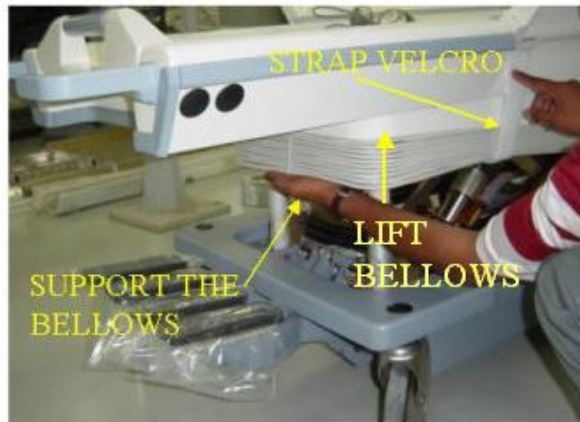
**CAUTION!!**

**DO NOT USE ALLEN KEY TO LOOSEN THE SCREW IT IS ONLY FOR HOLDING THE SCREW AND LOOSEN THE NUT BY USING A SPANNER AS SHOWN. IF YOU TRY TO LOOSEN THE SCREW BY ALLEN KEY IT MAY DAMAGE THE HEAD OF THE SCREW.**



**ILLUSTRATION 5G**

- 2.1.7.** Lift the bellows and using the Velcro straps provided, strap it as shown in Illustration 5H. (The Velcro is located inside the bellows and has to be unfolded and then strapped. Refer illustration 5I).



**ILLUSTRATION 5H**



**ILLUSTRATION 5 I**



2.1.8. Strap the bellows on all sides as shown. Refer illustration 6



**FASTENING OF BELLOWS TO SERVICE POSITION**

**ILLUSTRATION 6.**



**CAUTION!**

**DO NOT MOVE THE TABLE UP/DOWN WHEN THE BELLOWS IS IN SERVICE POSITION**

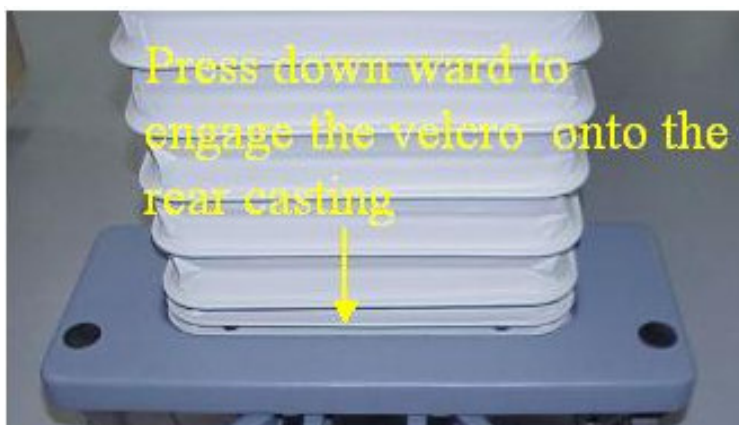
## 2.2. MOVING BELLOWS, TO WORKING/NORMAL POSITION.

- 2.2.1.** Unfasten all the Velcro straps and bring the bellows to bottom position. Make sure to fold back the Velcro straps and fasten inside as shown in illustration 7A. Otherwise the Velcro strap can get entangled while the Table is in up/down motion.



**ILLUSTRATION 7A**

- 2.2.2.** Bring the bellows down and fix the rear end of the bellows using the Velcro provided on the rear casting. Refer to illustration 7B.



**ILLUSTRATION 7B**

- 2.2.3.** Fix the two screws attaching the bellows on to the bellows strip and then onto Front casting. Refer to illustration 7C. For fixing these screws use an Allen key to hold Screw and tighten the nut below. Refer to illustration 7D

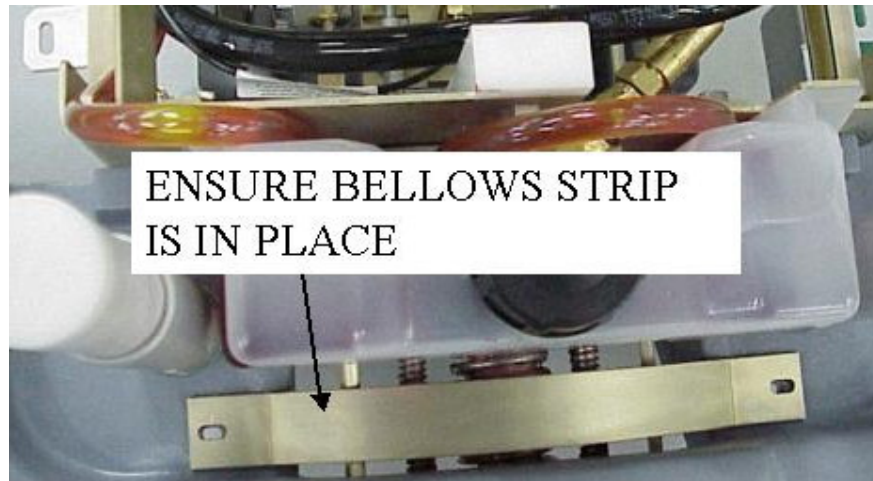


ILLUSTRATION 7C

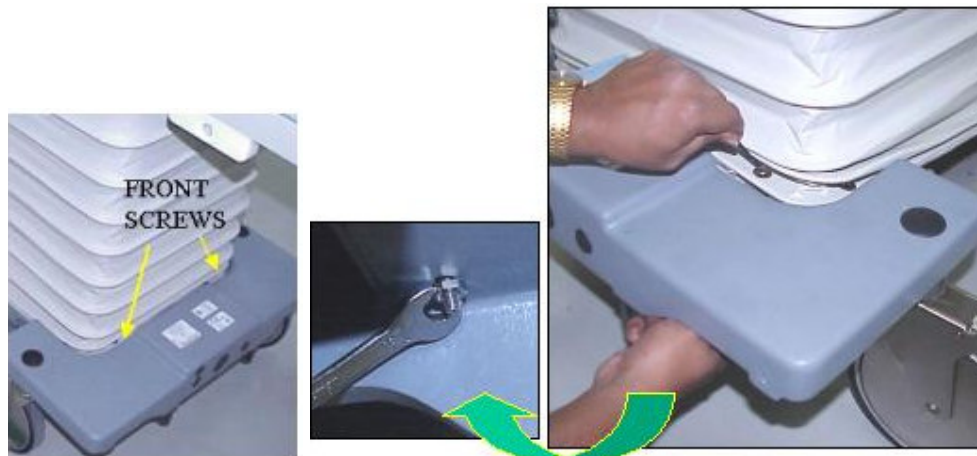


ILLUSTRATION 7D

- 2.2.4.** Attach the bellows onto the 3 brackets on either side. Refer illustration 7E. For Fixing these screws use an Allen key to hold the screw and tighten the nut below. Refer to illustration 7F.



ILLUSTRATION 7E

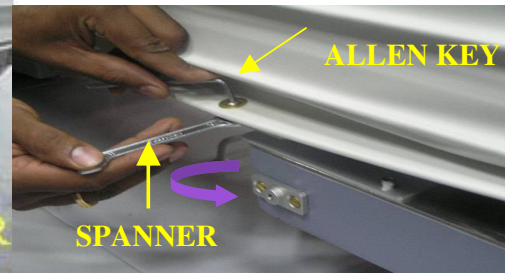


ILLUSTRATION 7F

- 2.2.5. Assemble the bottom cover using the locator on the casting as shown in illustration 7G and slide the bottom cover into position as shown in illustration 7H.



ILLUSTRATION 7G

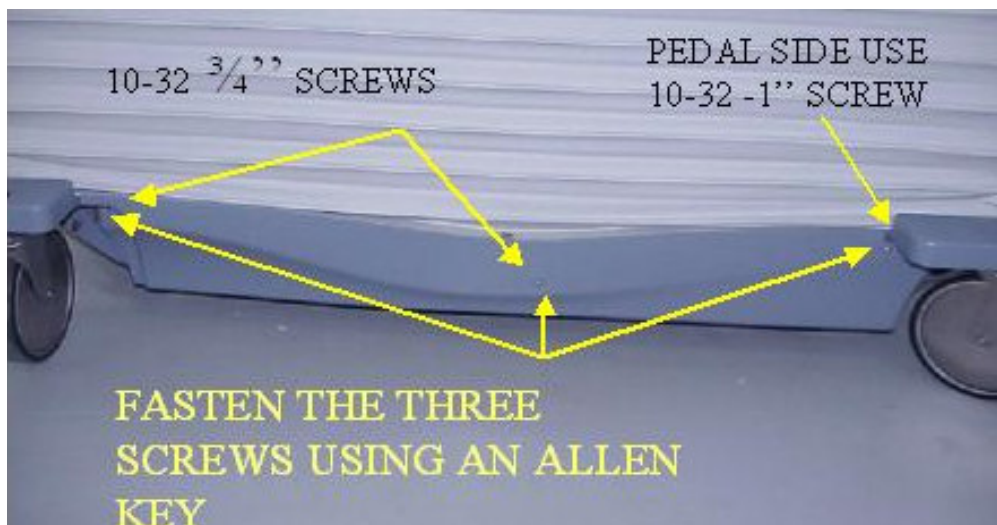


ILLUSTRATION 7H



**2.2.6.** Fix the bottom cover using 3 screws as shown. Refer illustration 7I.

Note: Middle screw to be engaged first. This will help in holding and locating the bottom cover in Position, then place the other 2 screws and tighten them. Refer illustration 7J, 7k and 7L.



**ILLUSTRATION 7i**



**ILLUSTRATION 7J**



**ILLUSTRATION 7K**



**ILLUSTRATION 7L**

### 3. LEVELING PATIENT TRANSPORT

#### 3.1. Required Tools

- 46-194427p255, Torque Wrench, ratchet type, 3/8 inch drive, Torque Range 30-200 in-lb.
- 46-307811p1 Socket Driver, hex Allen head, 5/32 inch bit, 3/8 inch drive.

#### 3.2. Checking Levelness of Patient Transport

- 3.2.1.** Patient Transport is leveled by adjusting caster height; however caster height must also be checked for uniform Loading.
- 3.2.2.** Dock the Patient transport. (Position the Patient transport in front of dock unit in case docking mechanism is not yet properly adjusted).
- 3.2.3.** Lock the caster brakes.
- 3.2.4.** With tabletop in fully raised position, check Patient Table for levelness, both front to back and left-to-right. This should be checked using a spirit level at the tabletop level, not at the table base, since tabletop alignment is the critical factor. Refer Illustration 8.

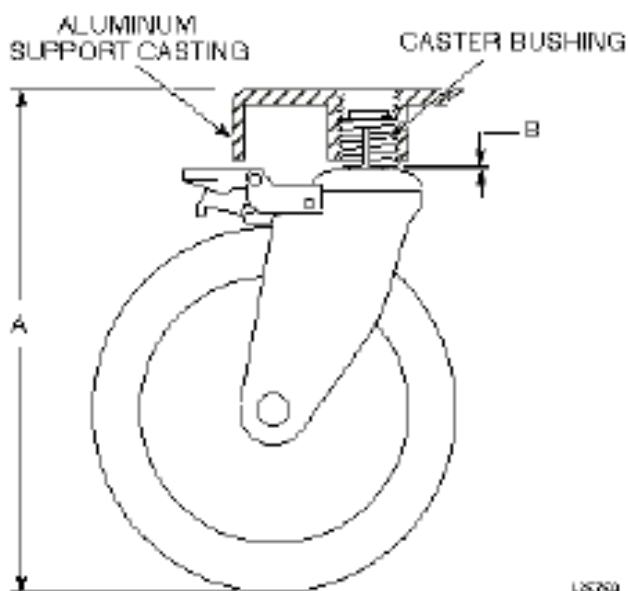


**CHECKING LEVELNESS OF PATIENT TABLE**  
**ILLUSTRATION 8**

- 3.2.5.** Check that all casters are sharing the load approximately equally by applying a small lifting force at each corner of the table and see if the caster easily lifts off the floor. If one or two diagonally mounted casters seem to be not loaded as much as the other casters, they will require adjustment.

### 3.3. Checking Caster Height Adjustment

- 3.3.1.** To keep track of measurement data, temporarily label each caster as 1, 2, 3, and 4. (Suggestion: label "steering" caster (green colored locking tab) as 1, and proceed labeling in a clockwise direction.)
- 3.3.2.** At each of the four caster wheel assemblies, measure the height from floor to top of Aluminum support casting (see Illustration 9, measurement A). Record this Measurement in Table 2, in MEASUREMENT A column, labeled 1ST HEIGHT. This height should be 11.0 inches  $\pm$  0.250 inches (279.4 mm  $\pm$  6.4 mm). If 1ST HEIGHT MEASUREMENT A is not within tolerance, calculate amount by which it is out of Tolerance ("+" if caster height is too small; "-" if height is too large) and record this value in 1ST DELTA column under MEASUREMENT A.



**CASTER MEASUREMENTS**  
**ILLUSTRATION 9**

TABLE 2

## MEASUREMENT A

## MEASUREMENT B

| CASTER | 1 <sup>ST</sup> HEIGHT | DELTA +/- | 2 <sup>ND</sup> HEIGHT | DELTA +/- | GAP | DELTA +/- |
|--------|------------------------|-----------|------------------------|-----------|-----|-----------|
| 1      |                        |           |                        |           |     |           |
| 2      |                        |           |                        |           |     |           |
| 3      |                        |           |                        |           |     |           |
| 4      |                        |           |                        |           |     |           |

**Note:** a) Tolerance for MEASUREMENT A: 11.0 +/- 0.250 inch.

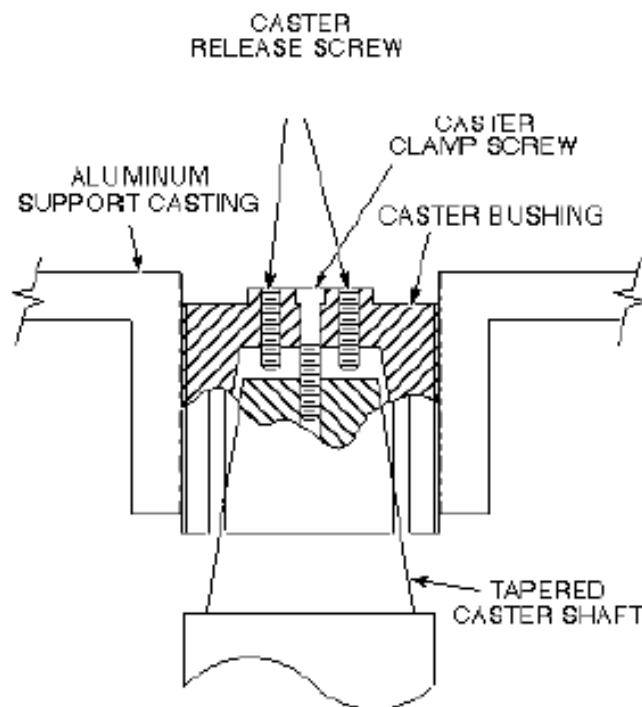
b) Tolerance for MEASUREMENT B: <0.250 inch.

- 3.3.3.** At each of the four caster wheel assemblies, measure the gap between bottoms of caster bushing and base of caster shaft (see Illustration 9, measurement B). Record this measurement in Table 2, in MEASUREMENT B column labeled GAP. The acceptable measurement for this gap is less than 0.250 inches (6.4 mm). If gap is not within tolerance, calculate amount by which it is out of tolerance ("+" if gap is too small; "-" if gap is too large, and record this value in DELTA column under MEASUREMENT B.

### 3.4. Caster Mounting Hardware Design Description

When the caster bushing can turn freely inside the aluminum support casting, the Bushing provides a means of adjusting caster height. The bushing is slotted and has a tapered hole to accept the tapered caster shaft. Refer Illustration 10. When the caster Clamp screw is inserted through the bushing and tightened securely into the caster Shaft, the shaft is drawn up into the bushing, causing the bushing to spread tightly Against the aluminum support casting threads, preventing the bushing from turning And changing the height adjustment. To adjust caster height, the clamp screw is backed out of the caster shaft, and the caster release screws are turned into the Caster bushing. The release screws push the caster shaft out of its tapered fit in the Caster bushing, leaving the bushing loose enough to turn and provide caster height adjustment.

After adjustment, the release screws must be backed out and removed to prevent contact with the caster shaft. Then the clamp screw can be adequately tightened, securing the caster shaft and immobilizing the bushing.



### CASTER MOUNTING CONSTRUCTION

#### ILLUSTRATION 10

There are two conditions that could cause the caster assembly to come loose and possibly fall off the Patient transport. These two conditions, their causes, and solutions are described below.

### 3.5. Problem Conditions and Causes

**3.5.1. The tapered caster shaft can become loose and back out of the caster bushing.** This condition can occur if the caster clamp screw has not been adequately tightened or is missing altogether. This condition can also be caused if the caster release screws were put in too far, before tightening the caster clamp screw, while adjusting caster height. The screws would be pushing against the caster shaft, preventing the shaft from seating securely into the bushing.

**3.5.2. The caster bushing can back out (partially or completely) of its threaded hole in the support casting.** If the tapered caster shaft becomes loose (as described in condition 1), or if the caster bushing has been improperly positioned during caster height adjustment (correct bushing positioning will be detailed in Section 3-6 Patient

transport Leveling Procedure) the bushing can back out of the aluminum support casting.

### 3.5.3. Solutions:

**3.5.3.1.** To avoid the condition caused by leaving the caster release screws in too far. Field engineer has to ensure the caster release screws are not accidentally left in position after height adjustment. The release screws should be used only When it becomes necessary to force the tapered caster shaft out of the caster bushing during an adjustment procedure, the two caster clamp screws will be used in the vacant caster release screw holes. One of the caster clamp screws will come from the caster being adjusted, and one will be borrowed from another caster assembly.

**3.5.3.2.** A **50 in-lb torque requirement** for the caster clamp screw. This torque on the caster clamp screw will result in sufficient tightness of the caster bushing inside the aluminum support casting to prevent the bushing from backing out when the caster assembly swivels.

## 3.6. Patient transport Leveling Procedure



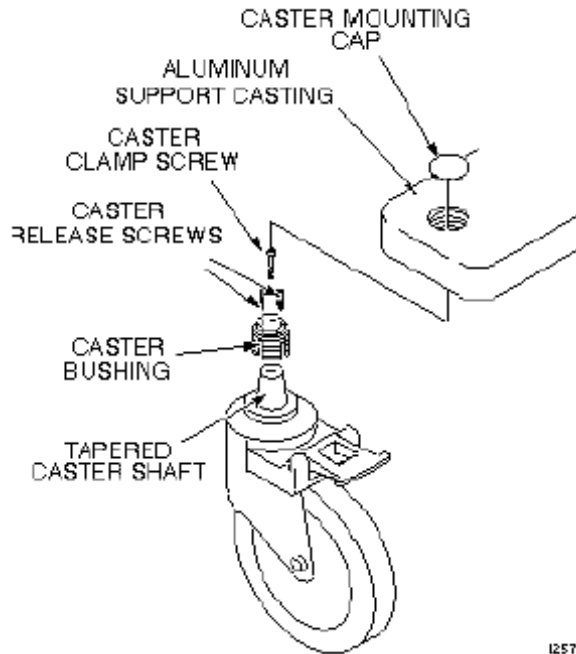
### **DANGER!!**

**FERROUS MATERIAL HAZARD!! TORQUE WRENCH AND SOCKET DRIVE CONTAIN FERROUS MATERIAL AND CAN BECOME FORCIBLY ATTRACTED TO THE MAGNET.DO NOT BRING THESE TOOLS INTO THE MAGNET ROOM.**

**3.6.1.** Undock Patient transport and move it out of magnet room.

**3.6.2.** On first caster to be adjusted, remove caster mounting cap by prying it off with a small screwdriver. See Illustration 11



**CASTER WHEEL ASSEMBLY****ILLUSTRATION 11****CAUTION!!**

**Adjustments that exceed tolerances given in the steps below leave Caster bushing vulnerable to damage, which can result in injury to patient and/or persons near Patient transport.**

- 3.6.3.** For each caster within tolerance, as determined in Table 2 proceed to step 3.6.14. of this Section. For each caster out of tolerance (i.e. has a 1ST DELTA entry in Table 2) adjust caster height by continuing with step 4.
- 3.6.4.** On first caster to be adjusted, remove caster clamp screw (cap screw) from its hole, and thread the screw into one of the empty caster release screw holes. Turn caster clamp screw in, finger tight only.
- 3.6.5.** Go to another caster assembly and remove the caster clamp screw. Bring caster clamp screw back to caster being adjusted.
- 3.6.6.** At caster being adjusted, turn second caster clamp screw into the other empty caster release screw hole, finger tight only.

- 3.6.7. Alternately screw in caster clamp screws one turn at a time until caster shaft is as loose as possible in caster bushing.
- 3.6.8. Using a means of leverage, slightly prop up aluminum support casting at caster being adjusted only enough to take Patient transport weight off caster wheel. This will Disengage caster shaft from caster bushing and prevent excessive wear on threads during adjustment.
- 3.6.9. Remove both caster clamp screws from the two caster release screw holes.
- 3.6.10. Adjust caster height an amount equal to the value recorded in 1ST DELTA column of MEASUREMENT A or MEASUREMENT B, whichever pertains, in Table 2. While holding caster wheel, use a 19 mm socket to turn caster bushing counter-clockwise to raise patient transport corner (for "+" DELTA), or clockwise to lower patient transport corner (for "-" DELTA). Each complete revolution of caster bushing raises or lowers caster 0.083 inches (2.1 mm).
- 3.6.11. Remove prop from under aluminum support casting.
- 3.6.12. Return borrowed caster clamp screw to its original caster assembly.
- 3.6.13. At caster being adjusted, insert remaining caster clamp screw into caster clamp screw hole.
- 3.6.14. Using the Torque Wrench (46-194427P255) and Socket Driver (46-307811P1), torque caster clamp screw to **50 inch-pounds**. If necessary, refer to operating instructions supplied with the torque wrench.
- 3.6.15. Verify that caster shaft is seated well into caster bushing and gap B is still within tolerance defined in Table 2.
- 3.6.16. Move Patient transport back into magnet room, and dock to the magnet.
- 3.6.17. For each caster, re-measure height A and record its value in Table 2, in MEASUREMENT A column labeled 2ND HEIGHT. If this 2ND value of A is out of tolerance, calculate amount by which it is out of tolerance, and record this amount in 2ND DELTA column under MEASUREMENT A.

- 3.6.18.** For each caster height still out of tolerance, adjust height as determined by value of 2ND DELTA in Table 2, by performing procedure beginning with step 3.6.4 of this Section.
- 3.6.19.** When all caster height measurements are within tolerance, replace all caster mounting caps.

**4. DOCK HOOK TENSION ADJUST**

- Attempt to dock the patient transport, paying attention to the force required to engage the docking mechanism.
- If the force to dock seems excessive, STOP and loosen hook (see next step).  
Verify that neither of the problem symptoms indicated in Table 3 is present.

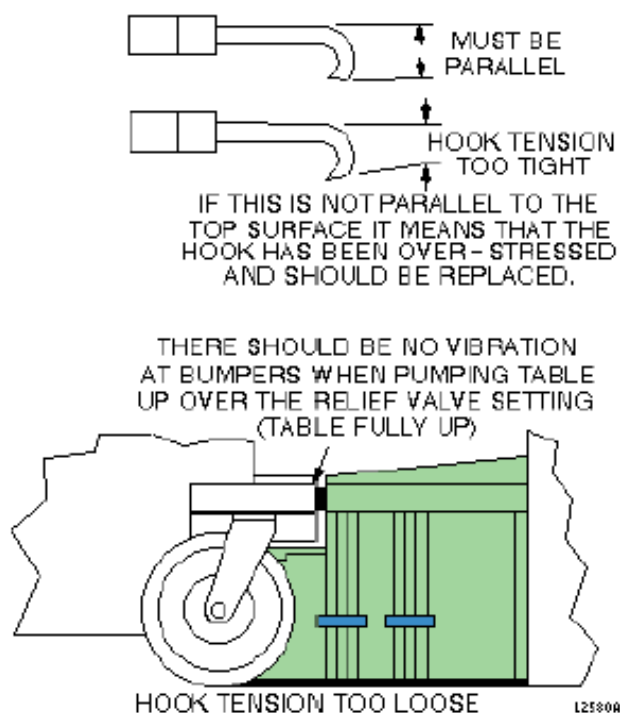
**4.1. DOCK HOOK MISADJUSTMENT SYMPTOMS****TABLE 3**

| <b>SYMPTOM</b>                                                                                                                                                                                | <b>CAUSE</b>           | <b>ACTION</b>                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|----------------------------------|
| Excessive force is required to depress dock pedal and fully engage dock latch. See Illustration 12                                                                                            | Hook tension too tight | Loosen hook one turn and retest  |
| Excessive back-and-forth vibration of patient transport when using "TABLE UP" foot pedal at dock. Especially Noticeable when tabletop is already in fully raised position See Illustration 12 | Hook tension too loose | Tighten hook one turn and retest |

**SYMPTOM CAUSE ACTION**

**Note:** Hook tension adjustment is very sensitive. Never adjust hook more than one turn at a time without re-testing.

- If dock hook tension requires adjusting, loosen setscrew and lock nut near base of dock hook, turn hook one turn, tighten setscrew and lock nut, and recheck hook adjustment. See Illustration 13.



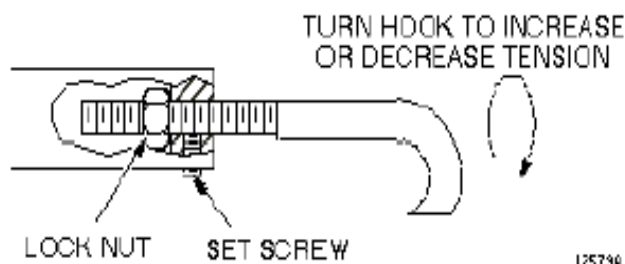
### DOCK HOOK TENSION CHECK

#### ILLUSTRATION 12



### CAUTION!!

**Do not adjust the dock hook tension more than one turn at a time. Adjusting more than one turn at a time may cause damage to internal Patient transport mechanisms.**



### DOCK HOOK TENSION ADJUST

#### ILLUSTRATION 13

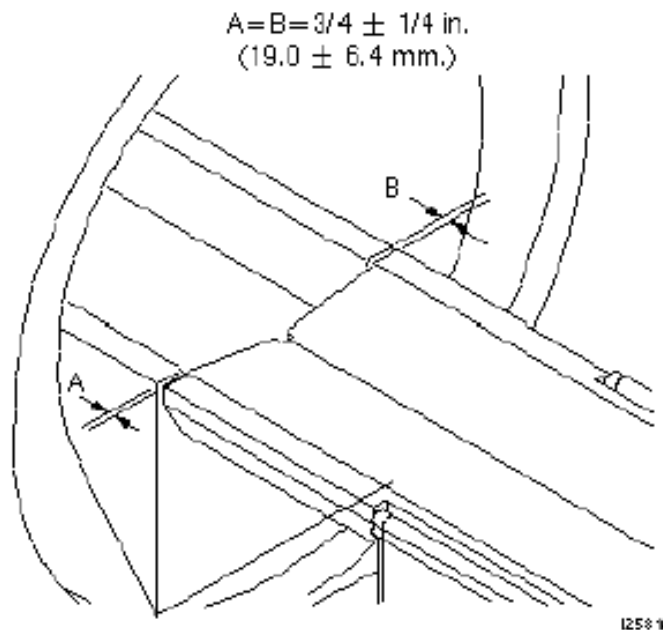
## 5. DOCK CENTERING

For proper Patient transport docking, the dock Assembly must be adjusted to meet two criteria:

- Gap between patient transport top and front of Magnet Enclosure must be  $3/4 \pm 1/4$  inch ( $19.0 \pm 6.4$  mm). This is to ensure that:
  - Gap is wide enough, so that there is no finger pinch hazard when docking patient transport
  - Gap is narrow enough so that UP LIMIT sensor operates properly
- Dock assembly must be square to front of magnet, so that the axis of cradle travel is the same for both patient transport and bridge.

### 5.1. Longitudinal Adjustment (i.e. in and out)

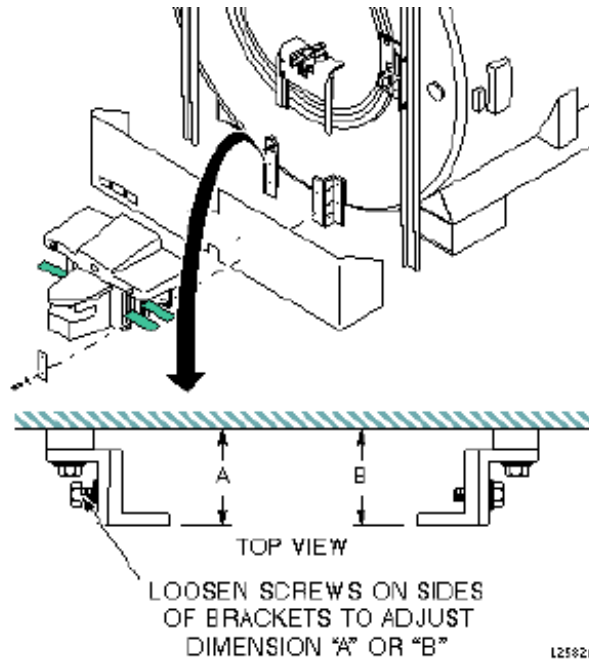
- 5.1.1. Measure distance from patient transport top to front cover. See Illustration 14 Dimensions A and B should be  $3/4 \pm 1/4$  inch ( $19.0 \pm 6.4$  mm) and should be equal to each other so the patient transport top is collinear with the bridge.



**PATIENT TRANSPORT DOCKING MEASUREMENT FOR DOCK**  
**ILLUSTRATION 14**

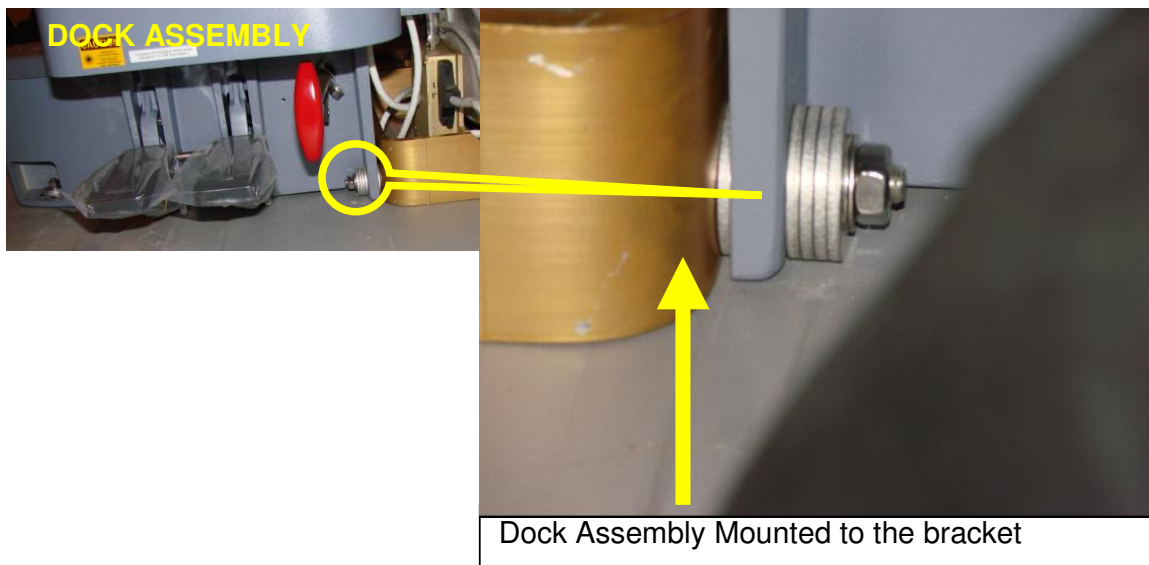


- 5.1.2. If adjustment is required, loosen screws on dock support brackets, move dock to proper position, and tighten screws. See Illustration 15 for location of dock support brackets.
- 5.1.3. Use Washers to set the gap between the table and the front bridge, i.e move the dock assy. front or back to set Dimension "A" and "B"  
See Illustration 16 A



**DOCK POSITIONING ADJUSTMENT**

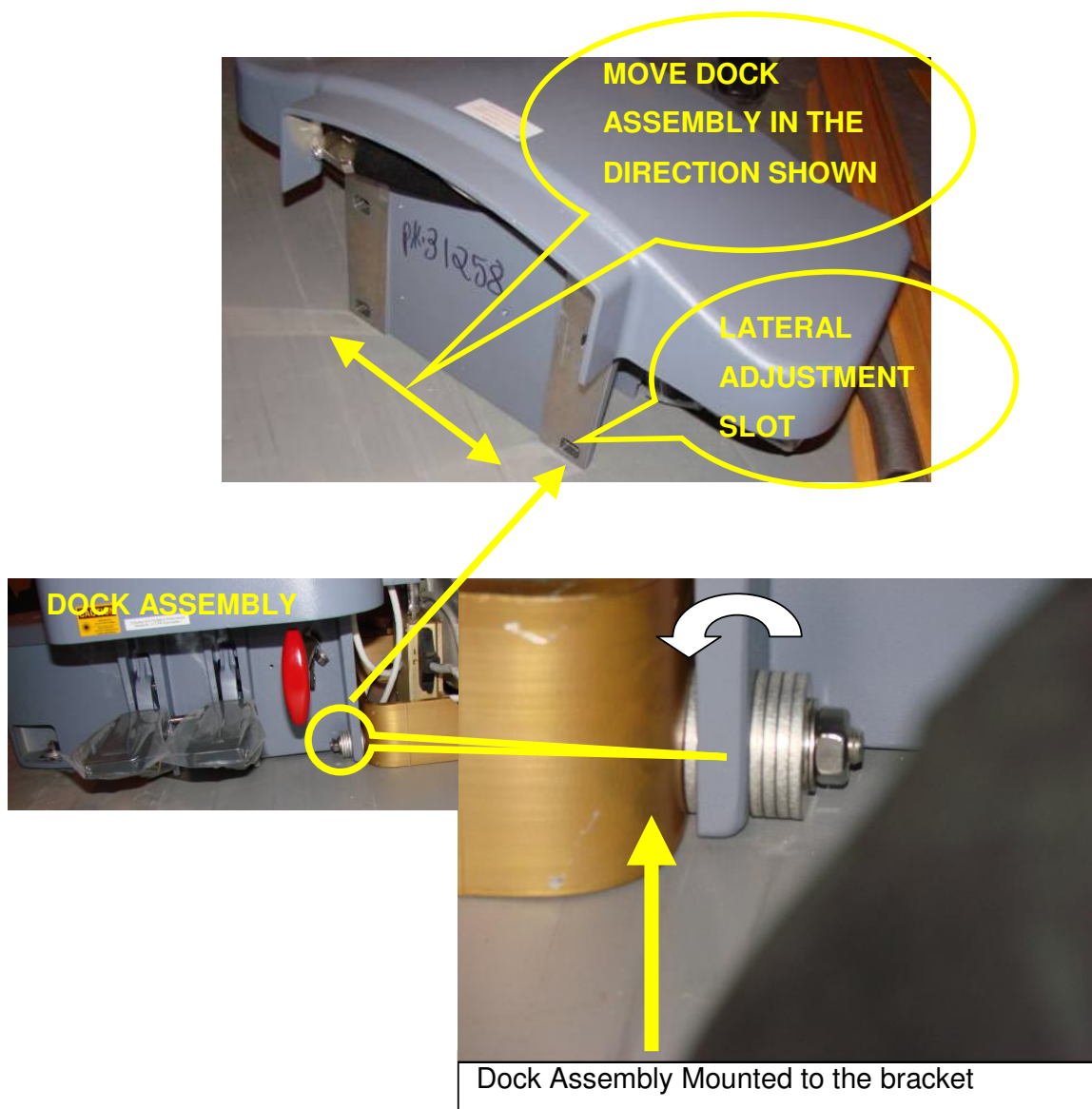
**ILLUSTRATION 15**



**ILLUSTRATION 16 A**

## 5.2. Lateral Adjustment (i.e. left to right)

- 5.2.1. Raise patient transport to maximum height using UP foot pedal.
- 5.2.2. Dock Patient transport.
- 5.2.3. Check alignment of Patient transport against bridge. Guide ridge down length of transport top should match guide ridge down length of bridge (left-to-right only, height has not yet been adjusted).
- 5.2.4. If adjustment is required, loosen screws on dock support brackets, move dock to proper position, and tighten screws. See Illustration 16B for location of dock support brackets



**ILLUSTRATION 16 B**

## 6. PATIENT TRANSPORT TOP HEIGHT ADJUSTMENT

### **DANGER!!**

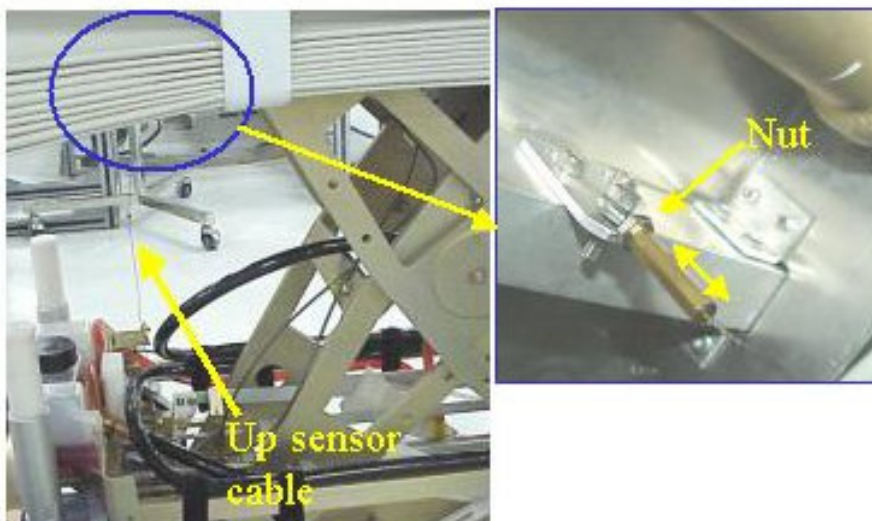
**FERROUS MATERIAL HAZARD!! TOOLS CONTAINING FERROUS MATERIAL CAN BECOME FORCIBLY ATTRACTED TO THE MAGNET. DO NOT BRING THESE TOOLS INTO THE MAGNET ROOM.**

### 6.1. TOOLS REQUIRED

- Spanner- 19 mm
- Nose Pliers

### 6.2. HEIGHT ADJUSTMENT

- 6.2.1. Dock Patient transport.
- 6.2.2. Measure the bridge height and make a note of the value.
- 6.2.3. Note the height difference between the table and the bridge (Table requires height adjustment, if there is a difference).
- 6.2.4. Next bring the patient transport out of the magnet room.
- 6.2.5. Remove bottom cover and move the Bellows to the top position as per the service procedure mentioned in [Section 2.1](#).
- 6.2.6. Bring the table down by 5 inches by pressing the down pedal, so that the scissors moves away from the dead stop.
- 6.2.7. Loosen the up-sensor cable by loosening the nut, using pliers and rotate the screw. Refer illustration 17.



**LOOSENING THE UPCABLE NUT**

**ILLUSTRATION17**

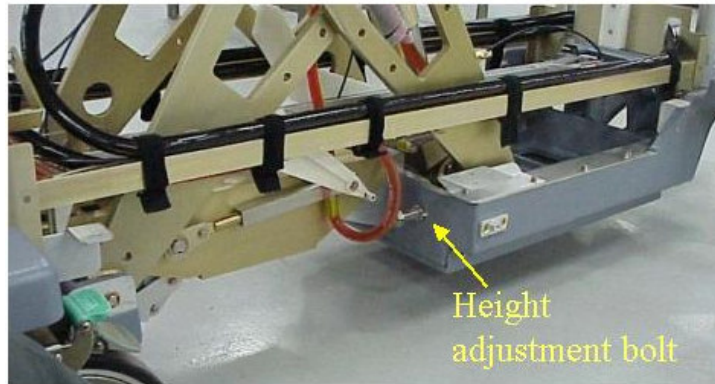
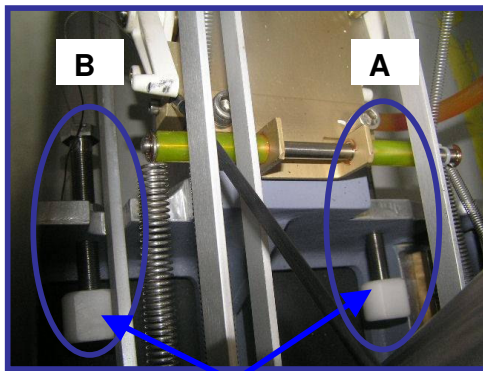
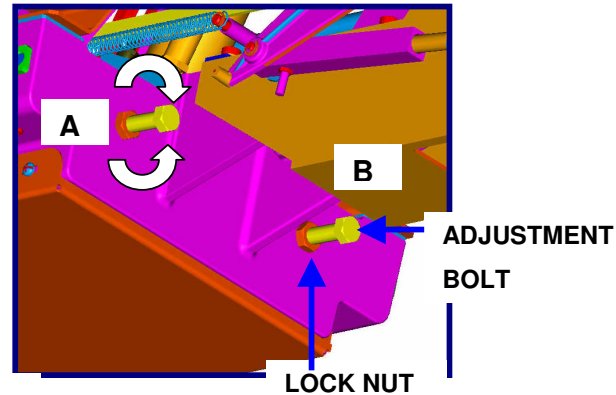


ILLUSTRATION 18A



DEAD STOPPERS



## ADJUSTING TABLE HEIGHT

ILLUSTRATION 18B

- 6.2.8 Loosen the lock nuts of both the dead stopper assembly A and B and use only one dead stopper assembly for height adjustment either A or B. Rotate the adjustment bolt clockwise to set the table at lower height and anticlockwise for setting the table at higher level for the chosen stopper assembly and Use a spanner to carefully adjust patient transport height by adjusting the adjustment bolt forward or backward as per the noted value in step 6.2.2. Refer Illustration 18A and 18B.
- 6.2.9 Pump the table up till the scissor mechanism flushes with the single dead stopper assembly. Refer illustration 19 and measure the table height, repeat above step if required to obtain the required table

height. After the required height is obtained tighten the locknut to lock the dead stop position.

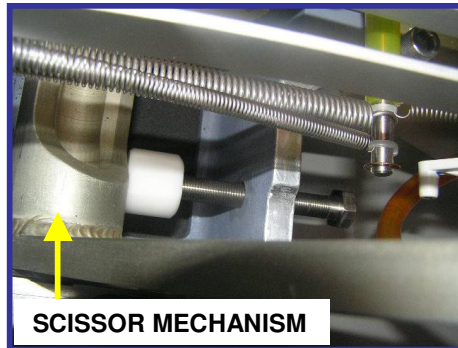


ILLUSTRATION 19

- 6.2.10 Pump the table up till the scissor mechanism flushes with the single dead stopper assembly. Refer illustration 19, in this position Rotate the adjustment bolt of other dead stopper assembly to make it flush with the Scissor mechanism. Refer illustration 20  
After ensuring both the dead stopper flushing with Scissor mechanism tighten the locking nuts of both dead stopper assembly

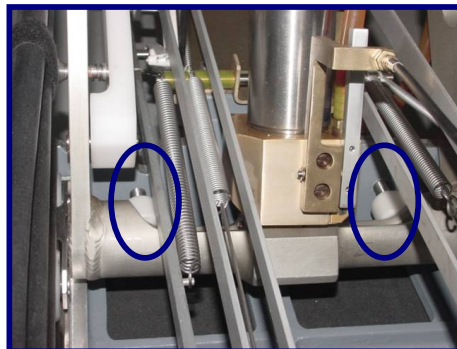


ILLUSTRATION 20



**CAUTION!!**

**Dual dead stopper assembly is available from Table Serial number: 30106WH9 introduced in HDe Table**



**ADJUSTING UPSENSOR CABLE****ILLUSTRATION 21**

- 6.2.11 Rotate the up sensor cable adjustment screw for fine-tuning and setting the top condition. Refer illustration 17 & 21. Rotate nut till cable is having no slackness.
- 6.2.12 Fix back the bottom cover using the 3 screws (Refer [Section 2.1](#)) and also bring the bellows back to normal working position and fasten it. Refer to [Section 2.2](#).

## 7. CRADLE RELEASE BLOCK ADJUSTMENT

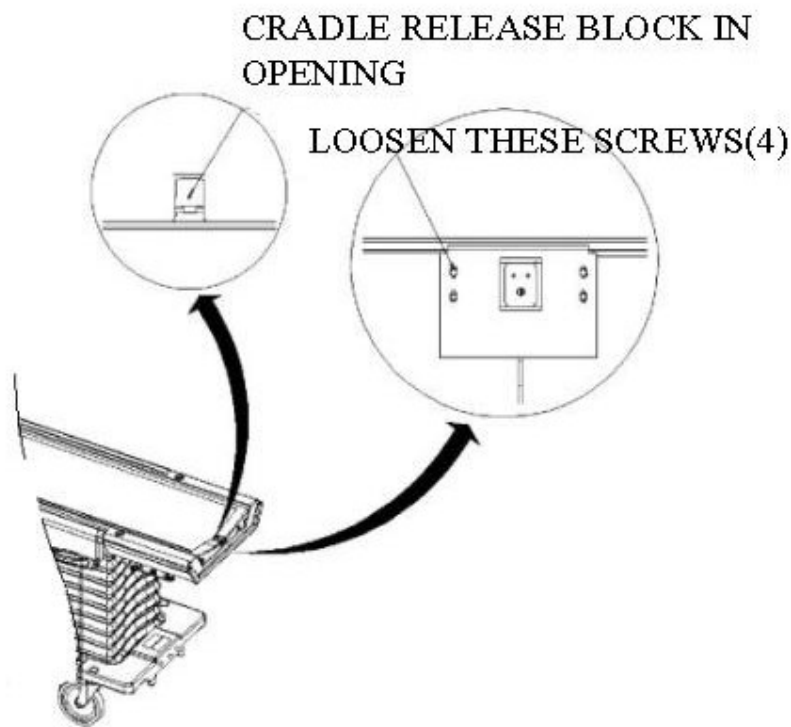


**DANGER!!**

**FERROUS MATERIAL HAZARD!! TOOLS CONTAINING FERROUS MATERIAL AND CAN BECOME FORCIBLY ATTRACTED TO THE MAGNET.DO NOT BRING THESE TOOLS INTO THE MAGNET ROOM**

### 7.1. Cradle Release Block Position Adjustment

- 7.1.1 Loosen four hex socket head screws securing the release block assembly. See Illustration 22.



### CRADLE RELEASE BLOCK POSITION ADJUSTMENT

#### ILLUSTRATION 22

- 7.1.2 Adjust the position of the assembly so that the release block is centered in the cutout on the front of the cradle assembly. Tighten hex socket screws. See Illustration 22.

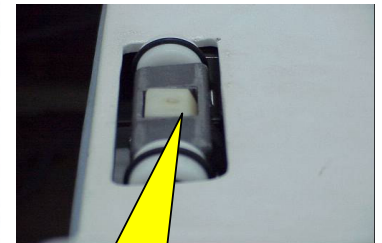


## 7.2. Cradle Release Block Retraction Adjustment

- 7.1.3 Dock Patient table.
- 7.1.4 Move the bellows to service position and fasten using the Velcro. Refer to [Section 2.1](#).
- 7.1.5 Loosen the lock nut, then tighten or loosen the cable adjustment screw in the pivot block so that the cradle release block is flush or just below the level of the casting in the tabletop. See Illustration 23A and 21B. And then tighten lock nut.



Illustration 23B



The Delrin block shall be flush or below the casting surface when in **docked** condition

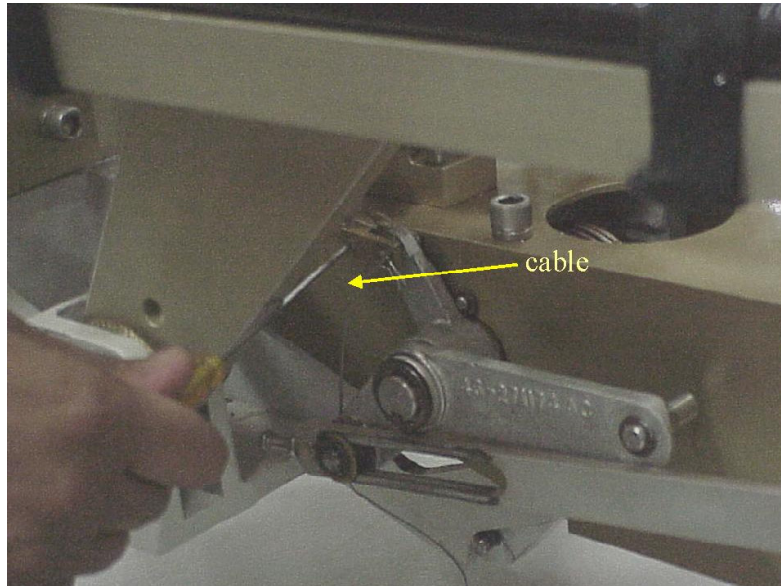
### CRADLE RELEASE BLOCK RETRACTION ADJUSTMENT

#### ILLUSTRATION 23A

**Note:** If the adjustment screw does not provide enough range for adjustment, coarse Adjustment may be made to the cable length where the cable end is clamped into actuator bracket. Fine adjustment will be required after performing a coarse adjustment.

For coarse adjustment follow the following procedure

- 7.1.6 Remove the setscrew as shown in illustration 24 below.
- 7.1.7 Remove the cable out and set cable length as required. Loop cable twice through the hole and clamp with setscrew.
- 7.1.8 Next follow the above steps 7.2.1-7.2.4 for fine adjustment.
- 7.1.9 Move bellows back to normal position. Refer [Section 2.2](#).



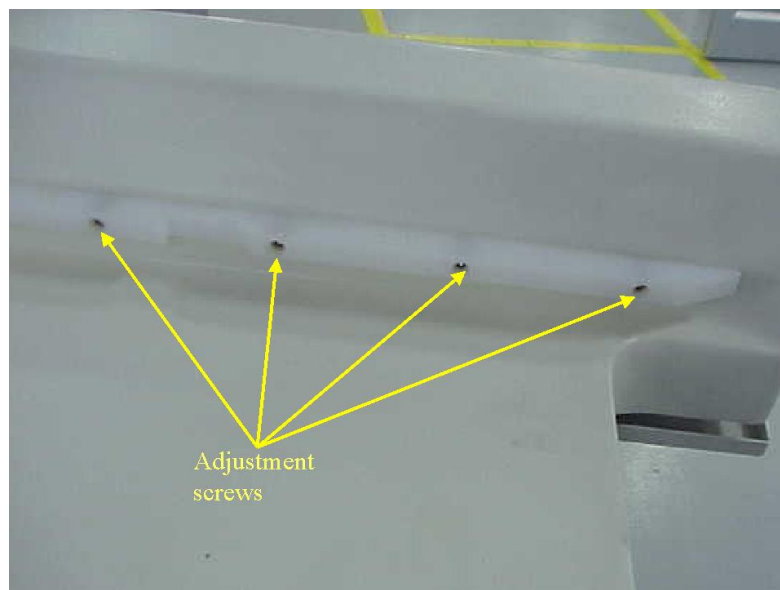
**REMOVING THE CRADLE RELEASE BLOCK CABLE**  
**ILLUSTRATION 24**

## 8. CRADLE GUIDE RAIL LATCH ADJUSTMENT

**Description:** The cradle is locked in home position on the Patient transport by a latch in the cradle which engages a notch in the cradle guide strip near the end of the Patient table. The cradle should be able to just engage this notch, without extra force, when pushed to home position on the table top, yet it must engage it tight enough so the table top (cradle) will not move on the Patient table when it is undocked.

### 8.1 Procedure:

- 8.1.1 Refer to procedure in [Section 1](#) for Removing Cradle Assembly, to partially remove cradle from tabletop.
- 8.1.2 Loosen five screws securing guide rail and slide guide rail forward or backward as required. Tighten screws and replace the cradle back. See Illustration 25.



**CRADLE GUIDE RAIL LATCH**  
**ILLUSTRATION 25**

## 9. CRADLE HOME/TABLE DOWN INTERLOCK ADJUSTMENT

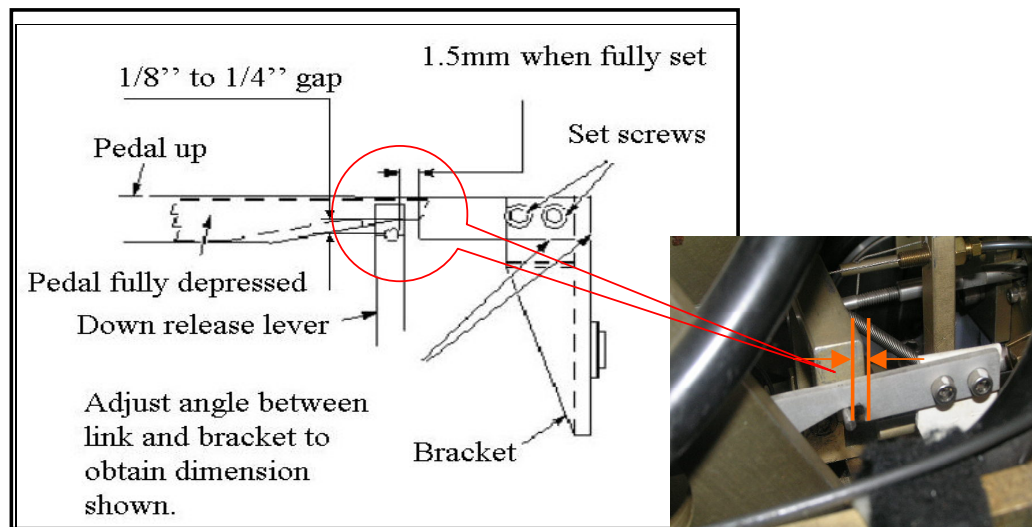
This adjustment is for the mechanical interlock, which prevents the Patient transport from being undocked or lowered while the cradle is in the magnet bore.

9.1 Move bellows to service position and fasten using Velcro straps. Refer to [Section 2.1](#).

9.2 Remove the cradle (Refer to [Section 1](#)), with cradle NOT HOME (removed), check the interlock mechanisms for tolerances shown in Illustration 26. To adjust:

9.2.1 (1.5 to 0.5 mm) gap - loosen the Setscrews shown in illustration 24 below. And adjust the angle between link and bracket to obtain dimension shown

9.2.2 The 1/8" to 1/4" The gap can be achieved by cable tension adjustment, for this follow the below procedure.



**CRADLE HOME/TABLE DOWN INTERLOCK**  
**ILLUSTRATION 26**

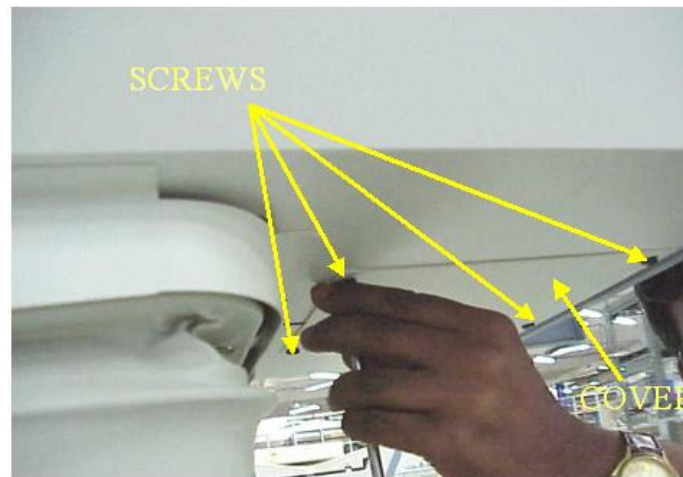
9.3 Remove the bottom hatchet as shown in the illustration 27A.

9.4 Using a spanner loosen the lock nut and rotate the adjustment screw outward till the gap mentioned above in step 9.2.2 is achieved. After setting the gap, tighten lock nut. See Illustration 27 B, 27C, 27D and 27E

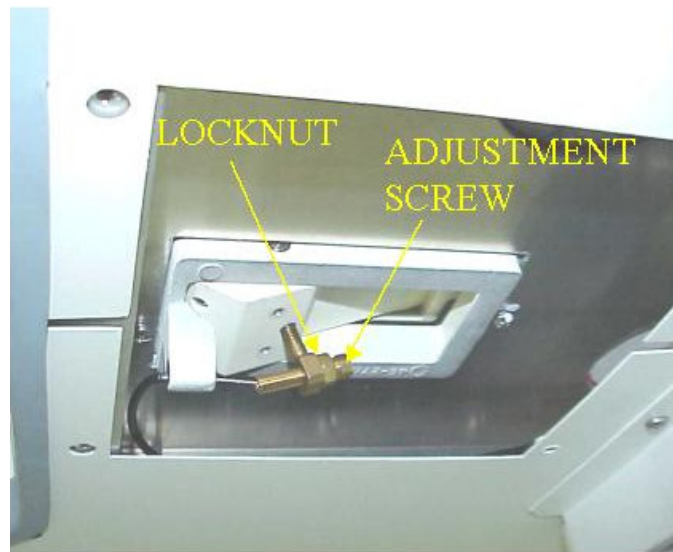
**Note:**

Illustration 27B & 27C is applicable for tables prior to execution of FMI - 60719J  
Refer illustration 27D & 27E For Table Serial starting from HDe Table Serial no:  
20521WH1&Signalite Serial No: 20477WH6 and after execution of FMI - 60719J

9.5 Place cradle back onto the table and move bellows back to normal working position. Refer to [Section 2.2](#).

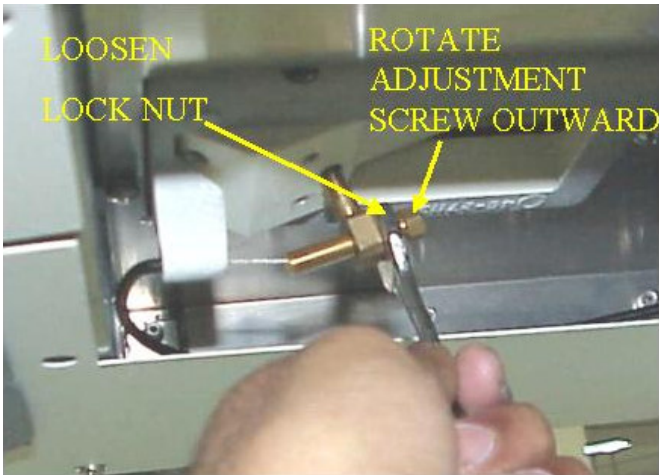


**REMOVAL OF HATCHET  
ILLUSTRATION 27 A**



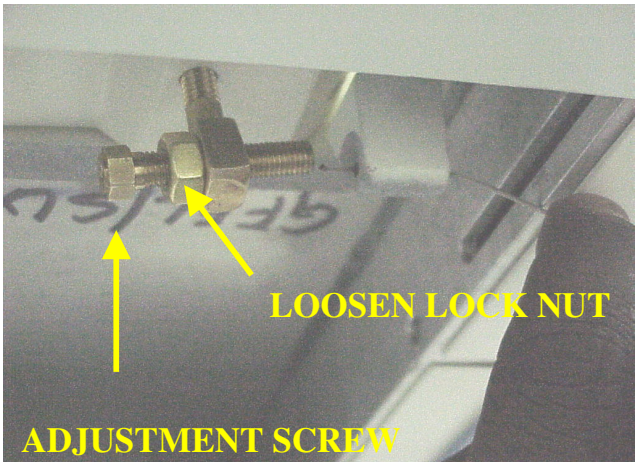
**CRADLE HOME/TABLE DOWN INTERLOCK ADJUSTMENT SCREW  
ILLUSTRATION 27 B**





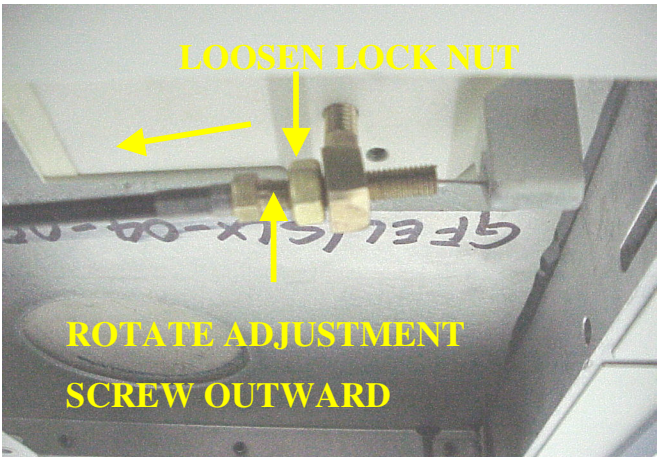
**ADJUSTMENT SCREW SETTING**

**ILLUSTRATION 27 C**



**CRADLE HOME/TABLE DOWN INTERLOCK ADJUSTMENT SCREW**

**ILLUSTRATION 27 D**



**ADJUSTMENT SCREW SETTING**

**ILLUSTRATION 27 E**

## 10. CRADLE RELEASE PIN HEIGHT ADJUSTMENT

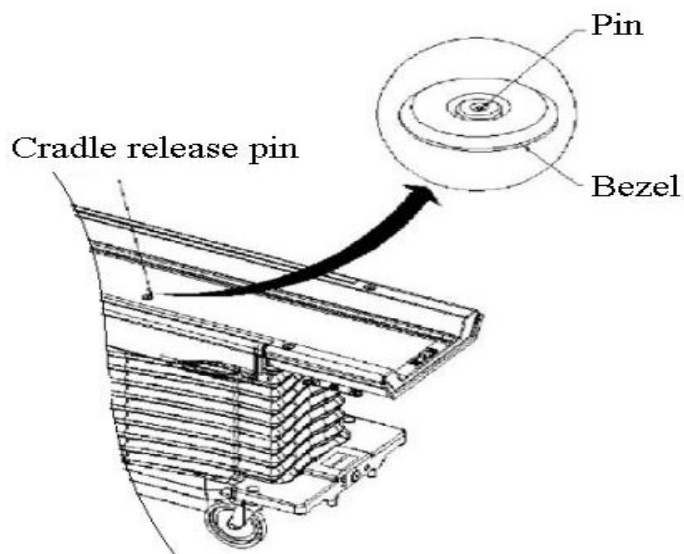
- 10.1 Remove Cradle Assembly from Patient table as per procedure in [Section 1](#).
- 10.2 Move bellows to service position and fasten using Velcro straps. Refer to [Section 2.1](#).
- 10.3 Tighten or loosen the cable adjustment screw so that the cradle release pin is **3mm** below the level of the tabletop. See Illustration 28 and Illustration 29A, 29B and 29 C
- 10.4 The Cradle pin height depends on the gap between the Table and Dock assembly. Ensure that the gap is about 3mm. See illustration 29D
- 10.5 Place cradle back onto the table and move the bellows back to normal working position. Refer to [Section 2.2](#).



**CRADLE RELEASE PIN HEIGHT ADJUSTMENT**

**ILLUSTRATION 28**

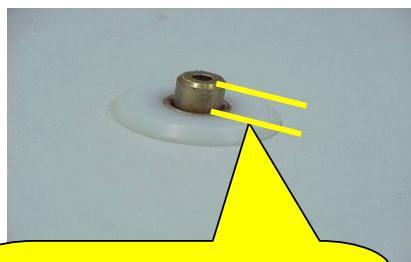




CRADLE RELEASE PIN POSITION

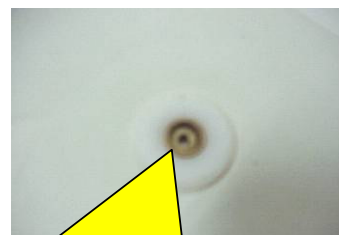
ILLUSTRATION 29A

Illustration 29B



The height of Cradle pin should be **4.7+/-0.25 mm** in **UNDocked** condition

Illustration 29C



The depth should be at least **3.0 mm** below brass top surface in **DOCKED** condition

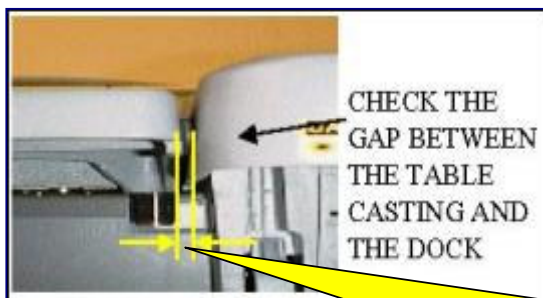
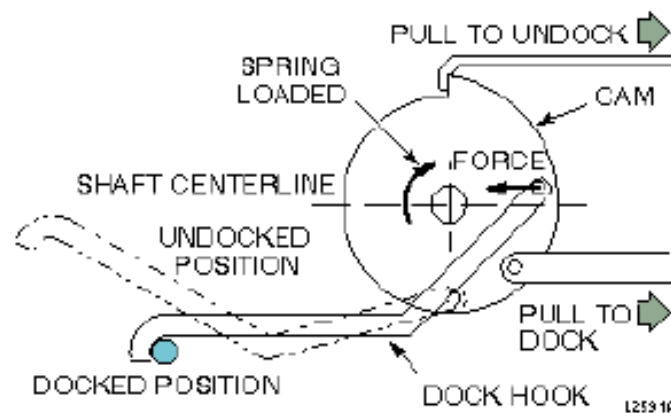


Illustration 29D

**NOTE:** The Cradle pin height depends on the gap between Dock assembly and Table. Ensure that the gap is about 3mm.

## 11. DOCK PEDAL STROKE ADJUSTMENT

Description: The table dock hook operates on the over-center principle. This is implemented by fixing one end of the hook to a cam on a shaft. When the force on the hook at its point of attachment to the cam is on the opposite side of the shaft as the hook end, this force will tend to keep the hook in the docked position. When the point of force is moved over-center of the shaft centerline, the force will no longer act to keep the hook retracted, and the spring loaded cam will extend the hook to the undocked position. See Illustration 30.



**DOCKING MECHANISM DIAGRAM**

**ILLUSTRATION 30**

### 11.1 Procedure

- 11.1.1 Move the bellows to service position and fasten using Velcro (Refer [Section 2.1](#)).
- 11.1.2 Tighten or loosen the stroke adjustment screw as required so that when the dock pedal is fully depressed during the docking operation, the hook cam mechanism just goes over-center. See Illustration 31.
- 11.1.3 Move the bellows back to normal working condition. Refer to [Section 2.2](#).

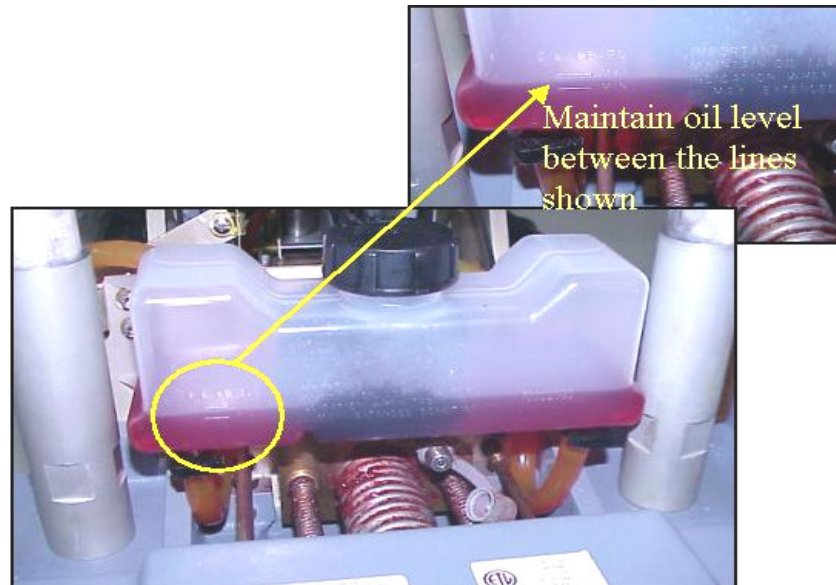


**DOCK PEDALSTROKE ADJUSTMENT**  
**ILLUSTRATION 31**

## 12. HYDRAULIC FLUID LEVEL CHECK AND FILL

**Note:** The below process varies for table serial No 2817WH5, 2818WH3, 2819WH1 which have an old design of reservoir, please refer to annexure 2 at the end of this manual for procedure for the same.

- 12.1 Move bellows to service/up position. (Refer [Section 2.1](#))
- 12.2 Raise table to FULL UP position.
- 12.3 Check hydraulic fluid level in reservoir at the oil level mark, Add fluid if necessary. By removing the cap See Illustration 32.
- 12.4 Move the bellows back to normal working condition. Refer to [Section 2.2](#)



**HYDRAULIC FLUID LEVEL CHECK**  
**ILLUSTRATION 32**

### 13. INSTALLATION OF HANDLE CAPS

The patient transport also has a feature of removable handles. This can be used In case of the customer having less space to install the table, in the magnet room, in such cases the table can be installed and docked to the magnet and the handle removed and the handle cap can be installed to close the handle cut out.

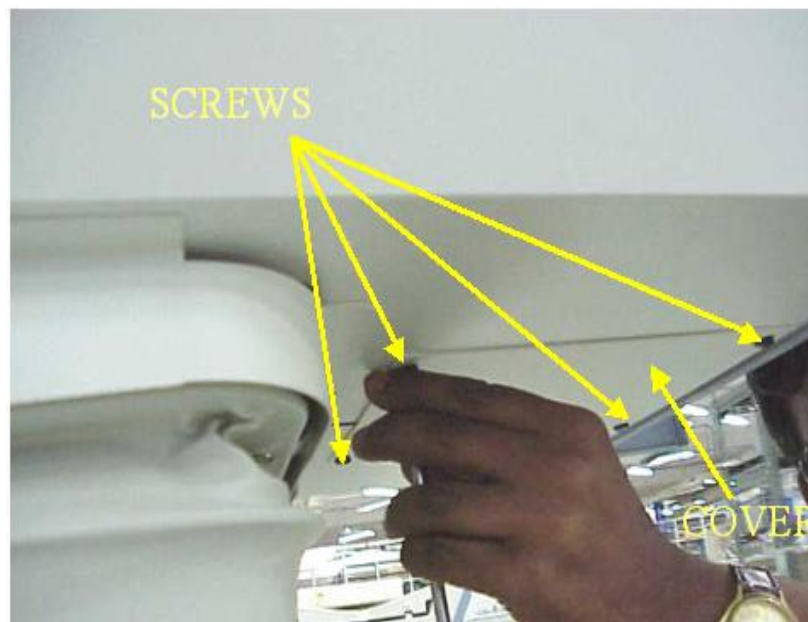
#### 13.1 Procedure:

13.1.1 The handle Caps is placed in a pouch in the hatchet below the table as shown.

Refer to illustration 33 and 33A. Remove the hatchet by removing the screws of the Hatchet, as shown.

13.1.2 Remove the caps LH and RH. And close the hatchet by fixing the screws. Refer to illustration 33.

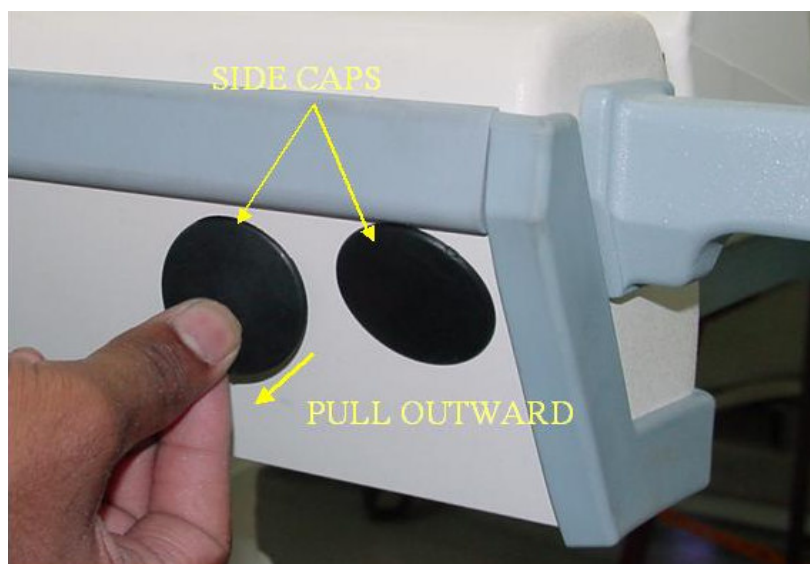
13.1.3 Remove the side caps as shown in the panel. Refer to illustration 34.



**REMOVAL OF HATCHET**  
**ILLUSTRATION 33**



**REMOVAL OF HATCHET**  
**ILLUSTRATION 33 A**



**REMOVAL OF SIDE CAPS**  
**ILLUSTRATION 34**

- 13.1.4 Remove two cap screws holding the handle as shown in illustration 35, and pull out the handle, Refer to illustration 36.





**REMOVAL OF CAP SCREWS**  
**ILLUSTRATION 35**



**REMOVAL OF HANDLE**  
**ILLUSTRATION 36**

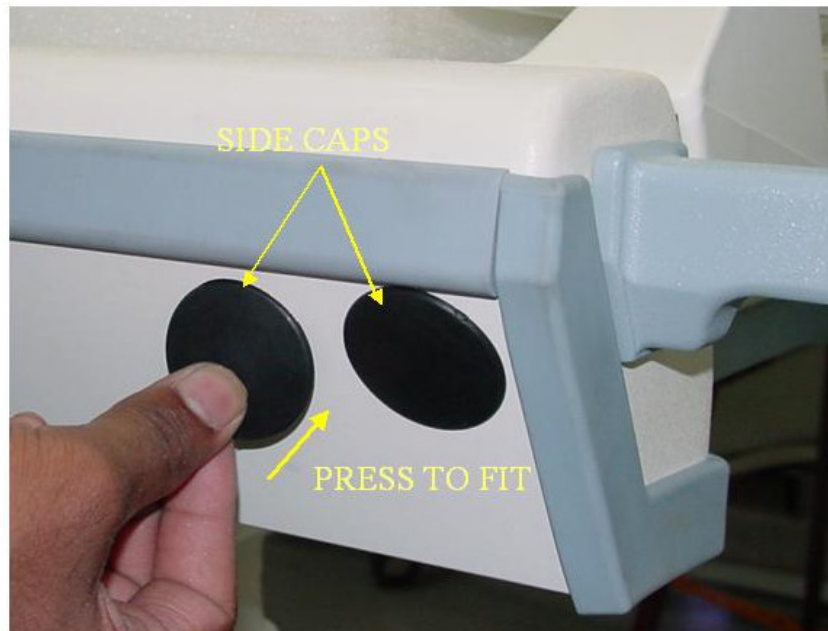
- 13.1.5 Close the cutout on the FRP where the handle was present using the handle Cap as shown. Refer Illustration 37. Repeat it for the other side handle. Care should be taken to fit the Respective Handle caps LH & RH.



**FITTING THE HANDLE CAP**

**ILLUSTRATION 37**

13.1.6 Fit back the side caps in the panel as shown. Refer to illustration 38.



**FITTING THE SIDE CAP BACK.**

**ILLUSTRATION 38**

## 14. DOWN RELEASE MECHANISM ADJUSTMENT

**Note:** If down pedal on the dockside is pressed and the down time is out of Specification of (24) sec, then refer to **Section 15** for adjustment of gap between dock unit and the down rod bumper.

**Note:** If the down pedal on the table is pressed and down time is out of spec (i.e. if the table takes longer to move down), then the down cable may be having slackness. To control the down time of the table and to remove any slackness in cable the following adjustment has to be made.

14.1 Move bellows to service position refer to [Section 2.1](#).



### CAUTION!!

**PERSONAL INJURY POSSIBILITY! WHEN PERFORMING MAINTENANCE ON THE TABLE IN ITS *UP* POSITION, ENSURE THAT THE SAFETY LOCK BAR IS SAFELY IN PLACE. SEVERE PERSONAL INJURY CAN OCCUR IF TABLE IS ACCIDENTALLY LOWERED DURING MAINTENANCE PERFORMANCE DO NOT LOAD THE TABLE WHEN THE TABLE IS UNDER MAINTENANCE.**

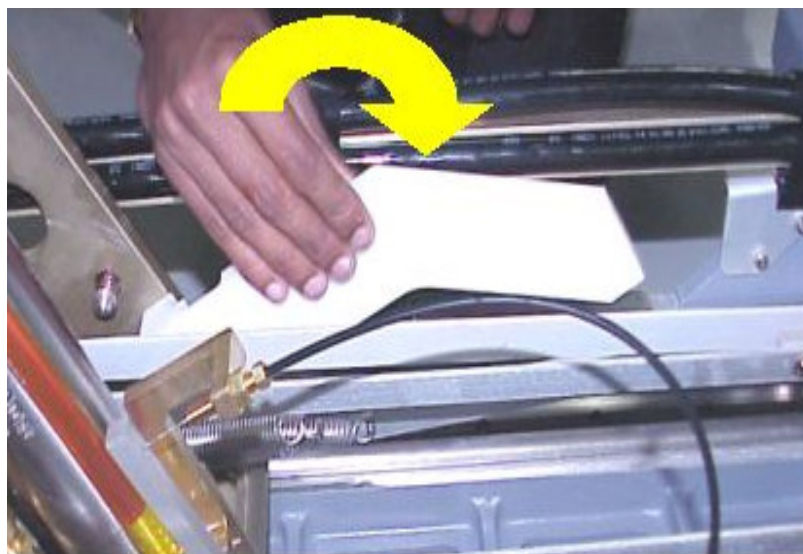
14.2 Engage the safety latch bar. By following the below steps.

14.3 Unhook Safety latch bar by pulling the lock pin knob outward as shown in the illustration 39A below, engage it towards the casting. Lower table by pressing the down pedal until lock bar fully supports tabletop. (Refer illustration 39B & 39C).



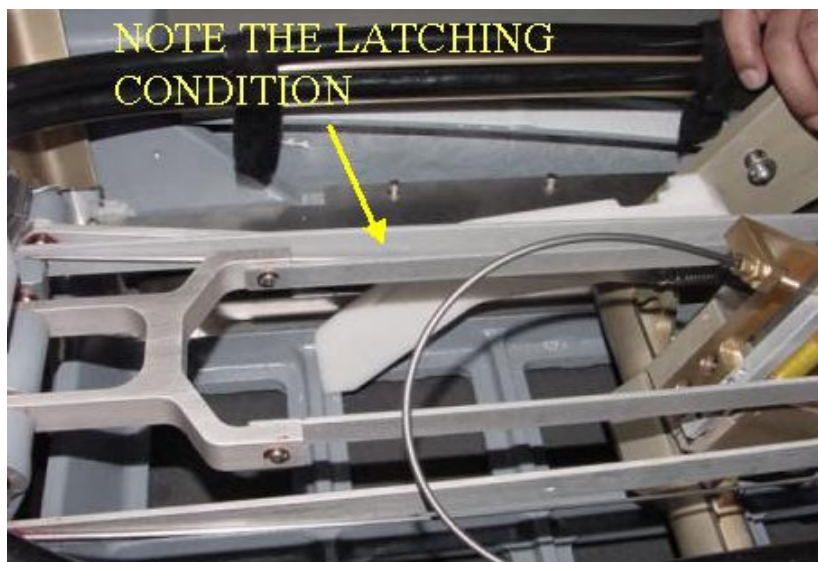
**UNLOCKING SAFETY LATCH BAR**

**ILLUSTRATION 39A**



**ENGAGING SAFETY LATCH BAR**

**ILLUSTRATION 39B**



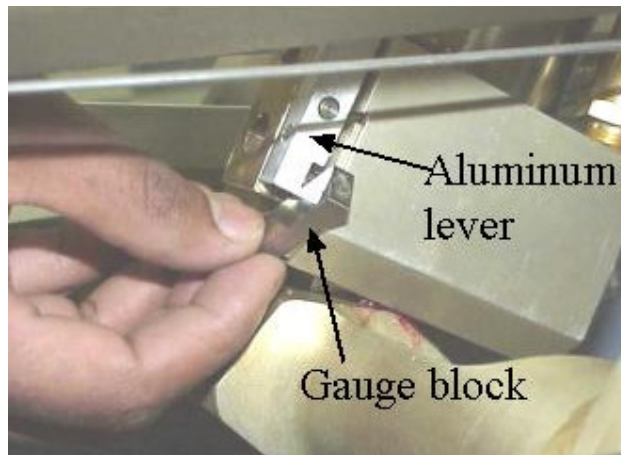
**ENGAGING SAFETY LATCH BAR**

**ILLUSTRATION 39C**

14.4 Place gauge (Part no. 5108543) below aluminum lever in assembly as shown.

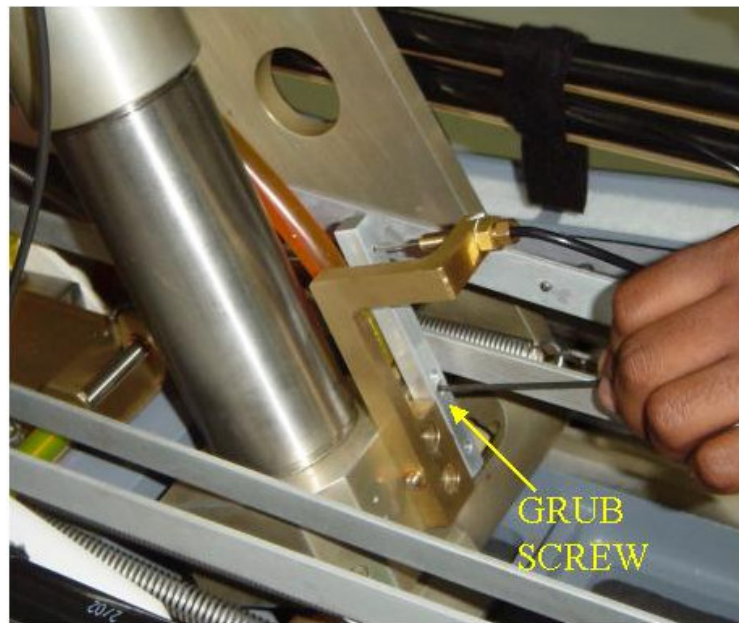
Refer illustration 39D





**ILLUSTRATION 39D**

- 14.5 Tighten grub screw on the aluminum lever. Refer illustration 39E below.



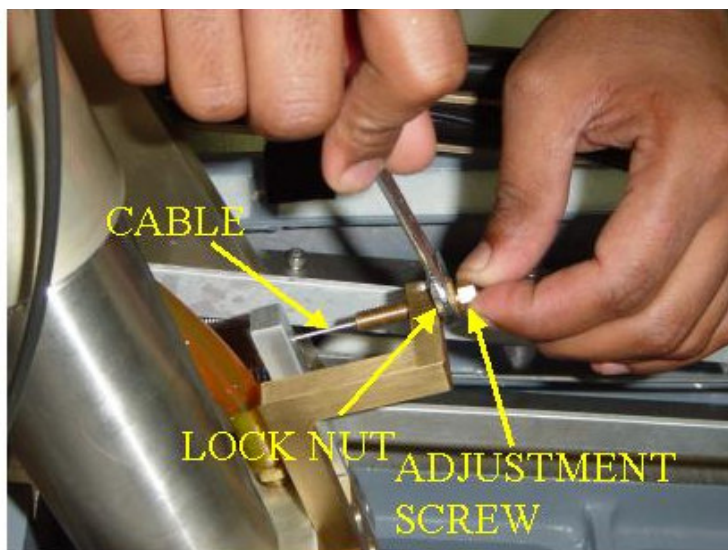
**ILLUSTRATION 39E**

- 14.6 Rotate adjustment screw "One rotation ANTICLOCKWISE " Refer Illustration 39F. Check if there is any slackness still, and if required rotate the adjustment screw further, till slackness in the cable is completely removed.



**ILLUSTRATION 39F.**

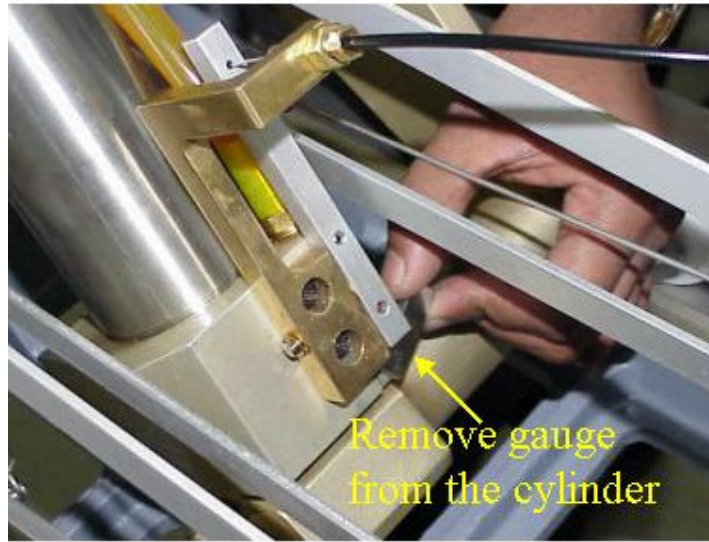
- 14.7 Hold the adjustment screw in this position and tighten the lock nut. Refer Illustration 39G.



**ILLUSTRATION 39G.**

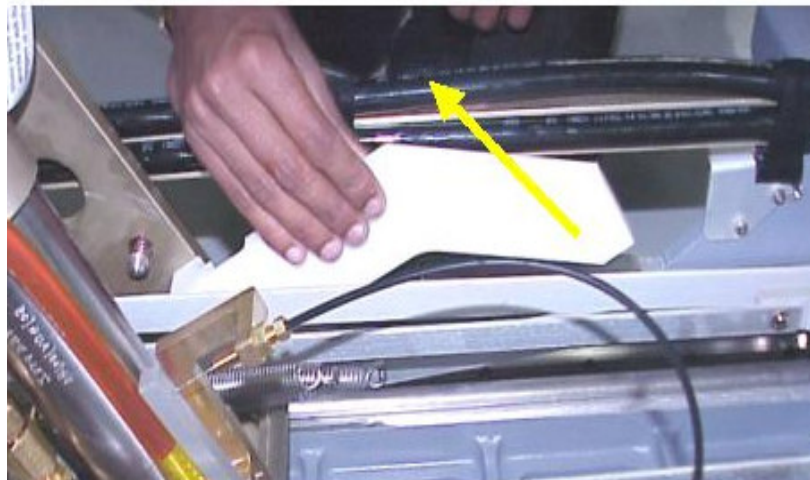
- 14.8 Remove the gauge from the cylinder. Refer illustration 40.





**ILLUSTRATION 40.**

- 14.9 Pull the safety latch bar away from the supporting rib of the casting and slide it back to its initial position, until it locks back in the spring lock provided (Refer to illustration 40A).
- 14.10 Ensure that the latch bar is firmly locked in its rest position. Refer to illustration 40B .



**ILLUSTRATION 40A.**



ILLUSTRATION 40B.

14.11 Check for normal functioning of the table from “Pedal side” as well as from “Dock side”, measure down time, if time is still more then the required specification of 24 sec and then repeat steps 14.1 to 14.11.

14.12 If the down time is within the specification, Replace bellows to normal condition. Refer to [Section 2.2](#).

**NOTE: Incase of replacement of cable or replacement of hydraulic cylinder then follow detailed instructions given in Section 21 of service replacement manual for setting of cable.**

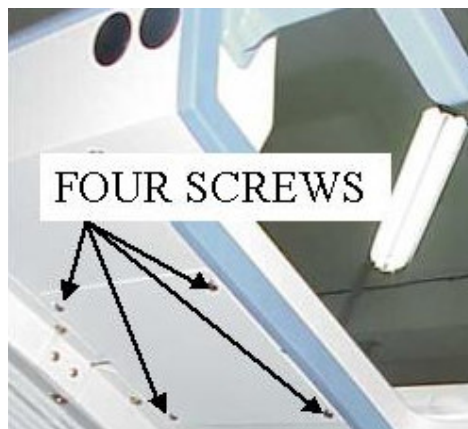
## 15. DOWN ROD GAP ADJUSTMENT

**Note: If down pedal on the dockside is pressed and the down time is out of Specification of (24) sec, then refer to section below for adjustment of gap between dock unit and the down rod bumper**

If the table down time is more than 24 sec when operated from the dock side pedal, then the gap between the down rod and the gantry casting has to be adjusted to control the down time of the table when operated from the dock end.

For this Washers have to be added on the down rod bumper.

- 15.1 The washers are sent with the table in a packet in the hatchet cover, retrieve the washers from the packet. Refer to illustration 41 below for removal of the hatchet.
- 15.2 Remove the four screws of the hatchet cover and carefully remove the cover plate, It contains a packet which contains the rubber cap for handle and also contains the washers for the down rod adjustment. Remove the washers from packet, and close the hatchet back by fixing the cover back by fastening the cover with the 4 screws. Refer to illustration 41&42.
- 15.3 Remove the 2 cap screws, which hold the front drip pan. Refer to illustration 43

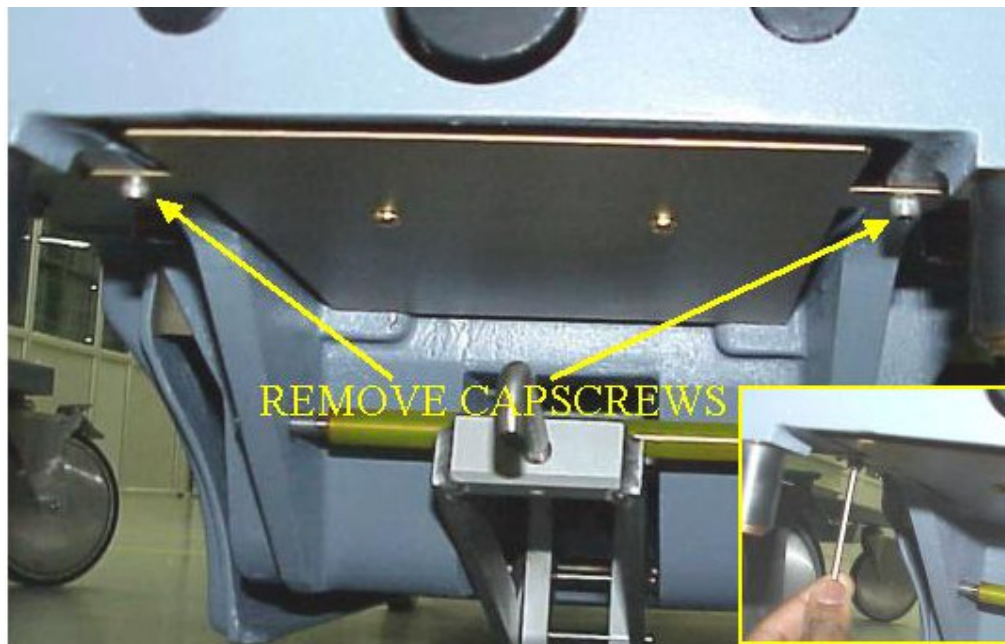


**REMOVAL OF HATCHET  
ILLUSTRATION 41**



REMOVAL OF SPARE WASHERS FROM HATCHET

ILLUSTRATION 42

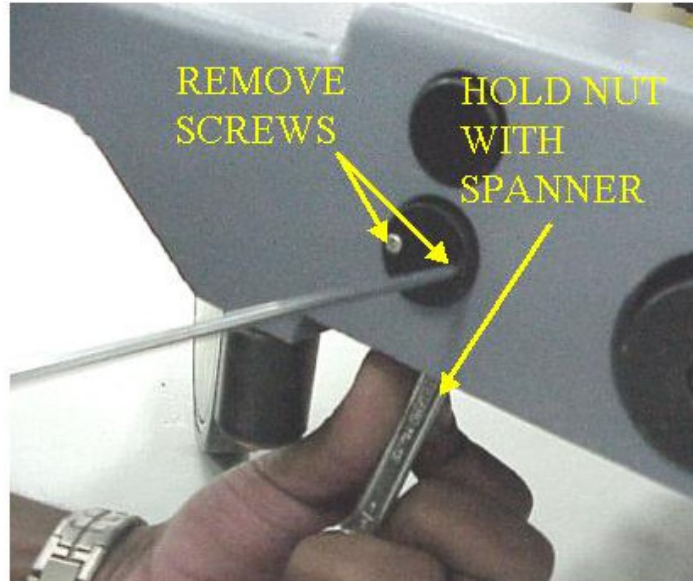


REMOVAL OF FRONT DRIP PAN

ILLUSTRATION 43



- 15.4 Using a spanner hold the nut connecting the down rod to the bumper and using a screwdriver remove the two screws from the bumper cap as shown in illustration 44 below.



**REMOVAL OF BUMPER CAP**

**ILLUSTRATION 44**

- 15.5 Add one washer to the Bumper Cap (Refer to illustration 45).



**ASSEMBLE THE WASHERS**

**ILLUSTRATION 45**

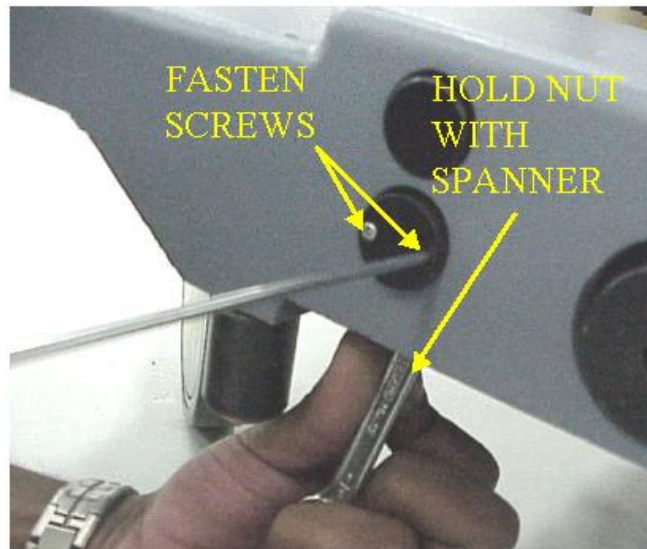
- 15.6 Insert two screws into any two holes provided as shown to make a sub assembly (Refer to illustration 46).



**INSERT THE SCREW TO MAKE A SUB-ASSEMBLY**

**ILLUSTRATION 46**

- 15.7 Insert this subassembly onto the bumper and hold the nut (fastening the bumper to the down rod) using a spanner and fasten the screws (use any two threaded holes diagonally positioned). Refer to illustration 47 below.



**FASTENING OF THE BUMPER CAP WITH WASHERS**

**ILLUSTRATION 47**

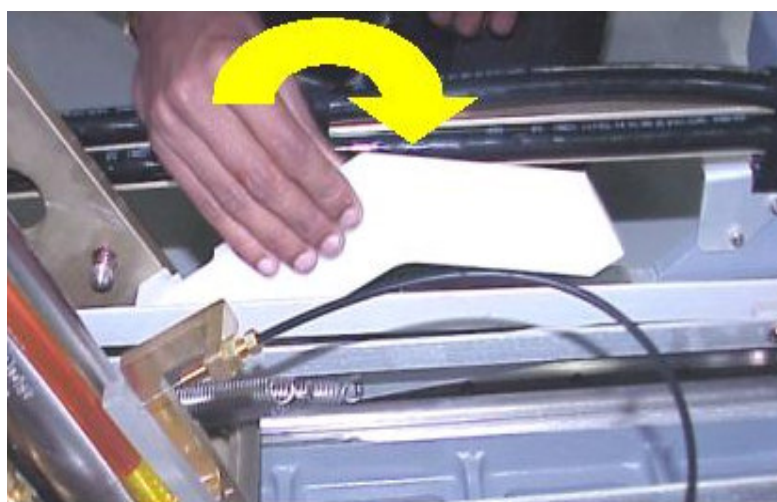
- 15.8 Move bellows up (Refer to [Section 2.1](#))
- 15.9 Engage safety latch by pulling the lock pin knob outward and engage it towards the casting, then lower table by pressing the down pedal until lock bar fully supports table top. Refer illustration 48, 49 and 50





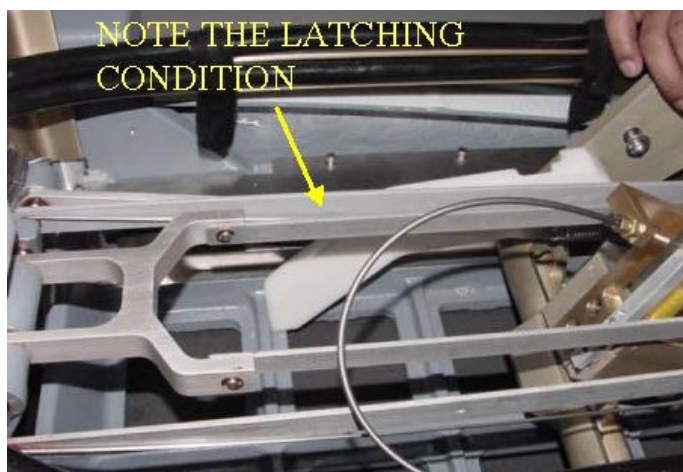
**UNLOCKING SAFETY LATCH BAR**

**ILLUSTRATION 48**



**ENGAGING SAFETY LATCH BAR**

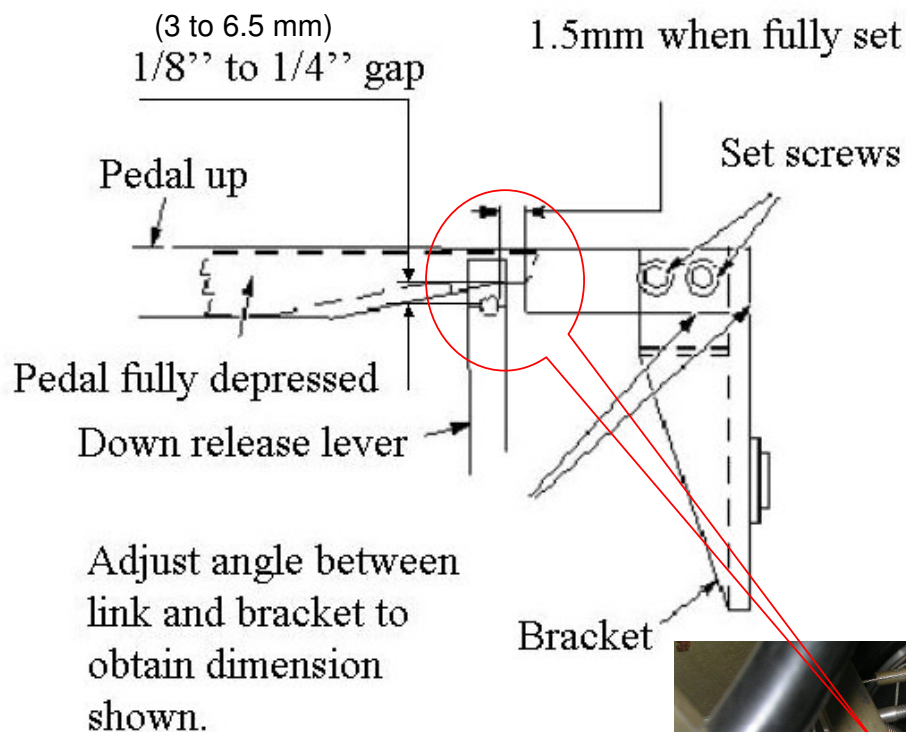
**ILLUSTRATION 49**



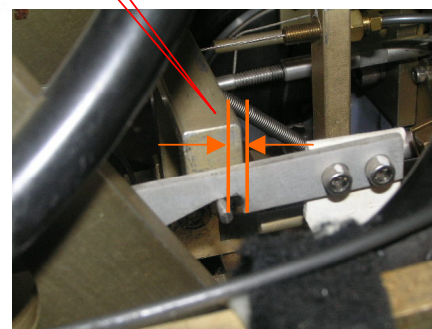
**ENGAGING SAFETY LATCH BAR**

**ILLUSTRATION 50**

- 15.10 Dock the table and check the gap between the down release lever and down link.  
Refer illustration 51. (It should be 1.5 mm. if it is less than 1.5 mm. Remove washer added in step 15.5 to achieve this gap).



**ILLUSTRATION 51**



**Note: If this gap (of 1.5mm) reduces and more washers are added on the Push rod, it leads to “spontaneous down” of Table.**

15.11 If the gap is 1.5 mm, disengage the safety latch bar. Refer illustration 52.

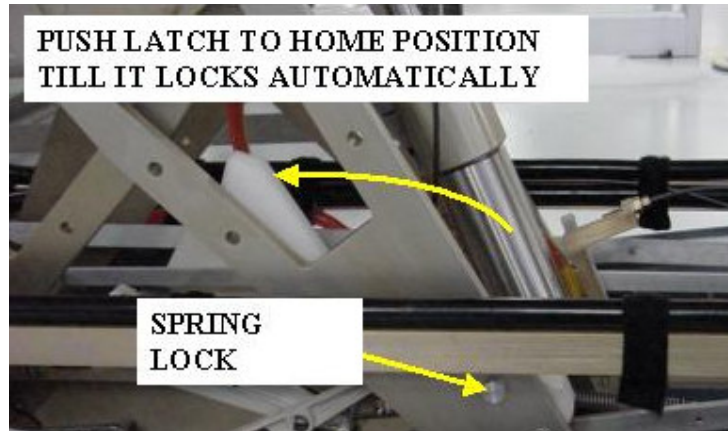


ILLUSTRATION 52

15.12 Check the down time, using the down pedal on the dock unit. If the time taken is still more than the specified time of 24 sec then add one more washer. Repeat steps 15.4 to 15.11. Again check the down time. If it is within specifications then, fix the Front drip pan back. Refer to illustration 43.

## 16. CRADLE SIDE LOCK ADJUSTMENT

**16.1** Undock patient table, and move it out of MR room.

**16.2** Invert the cradle and place on a flat surface

**16.3** Loosen the Delrin screw shown in Illustration 53.

**16.4** Remove the wheel housing nearest to the Delrin lock as shown in illustration 54

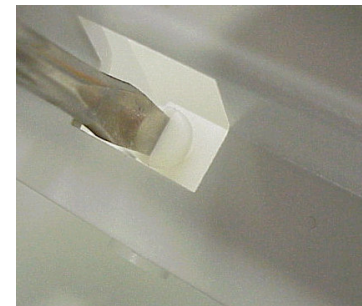
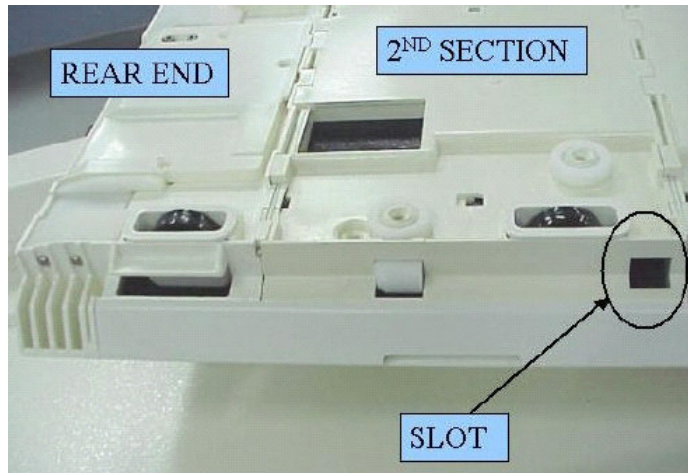


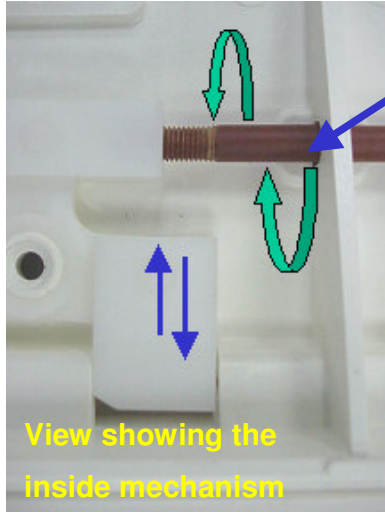
Illustration 53



Illustration 54

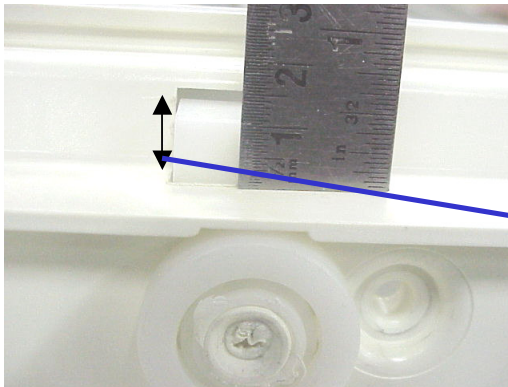


- 16.5** Insert Finger to reach the Push rod and rotate the push rod using finger, so that the Delrin lock length increases to 10mm. (refer illustration 55A and 55B)



PUSH ROD CAN BE ROTATED TO MOVE DELRIN LOCK OUTSIDE /OR INSIDE.

**Illustration 55A**



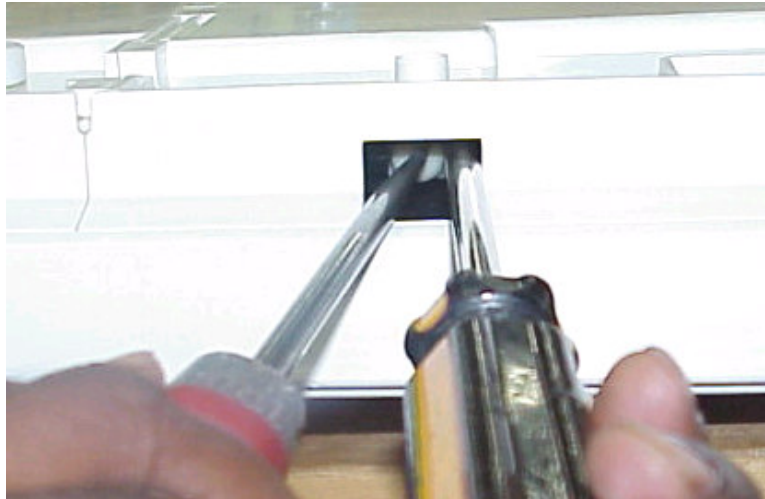
**Illustration 55B**

INCREASE THE DELRIN LOCK LENGTH TO 10MM.

- 16.6** Push the Delrin screw block to tension the tygon tube inside and under tension lock the Delrin screw. As shown in illustration 56&57.

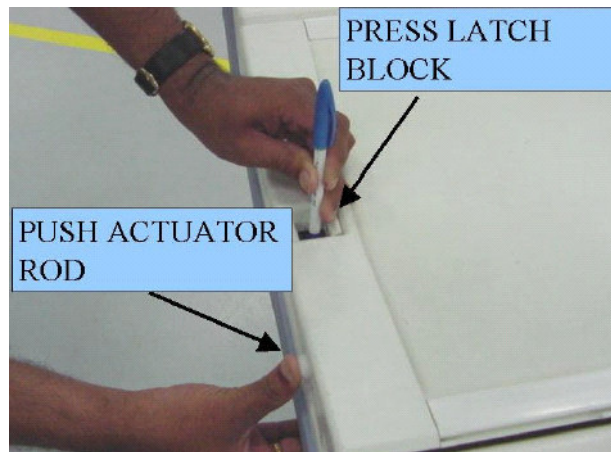


**Illustration 56**



**ILLUSTRATION 57**

- 16.7** Place cradle back on table and check if it is locking, by depressing the Push rod at front of cradle by 4 mm and slide cradle out and check if lock is releasing smoothly. Refer illustration 58.



**ILLUSTRATION 58**



## 17. Front guide Rollers Adjustment

17.1 Undock patient table, and move it out of MR room.

17.2 Remove the Cradle assembly as per Section 1

17.3 Front guide Rollers Height Adjustment:

Front guide rollers are to be at .25" above the cradle wheel tracks as shown in ILLUSTRATION 59

In case of any variation follow the next steps of adjustment process.



ROLLERS

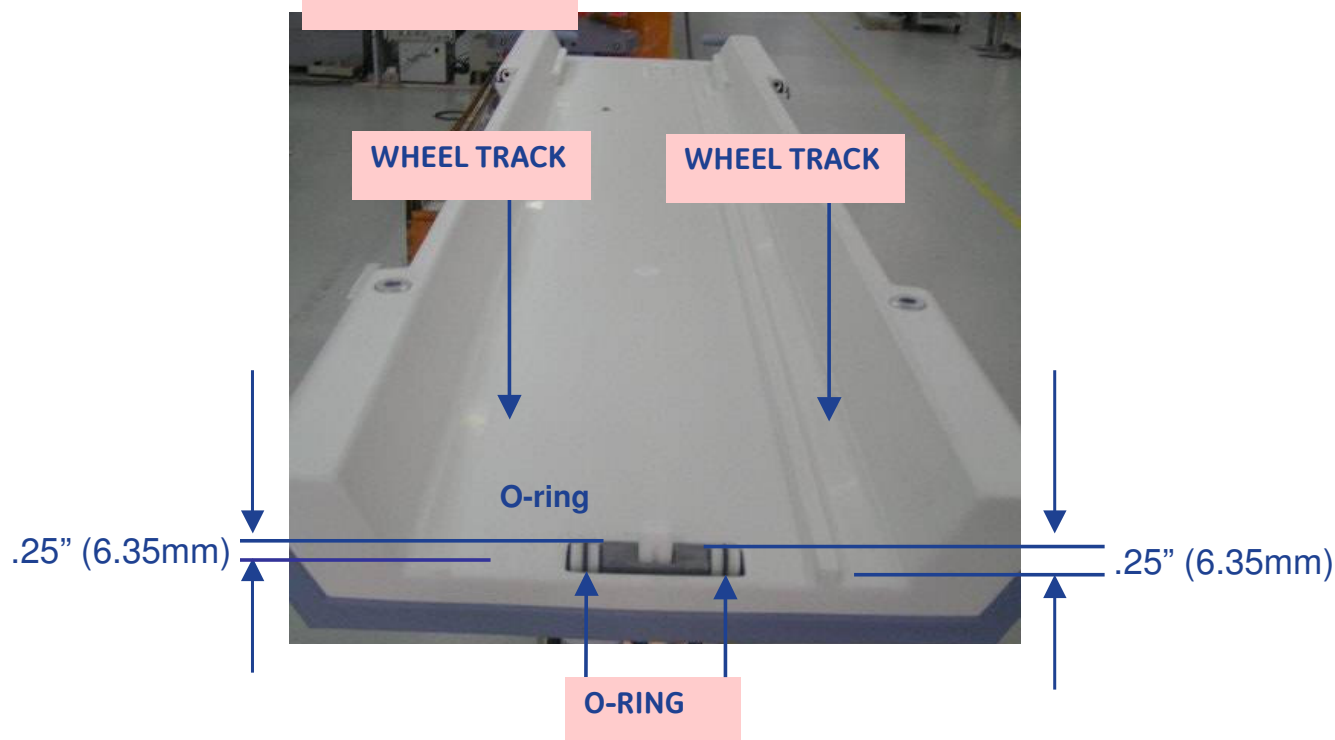
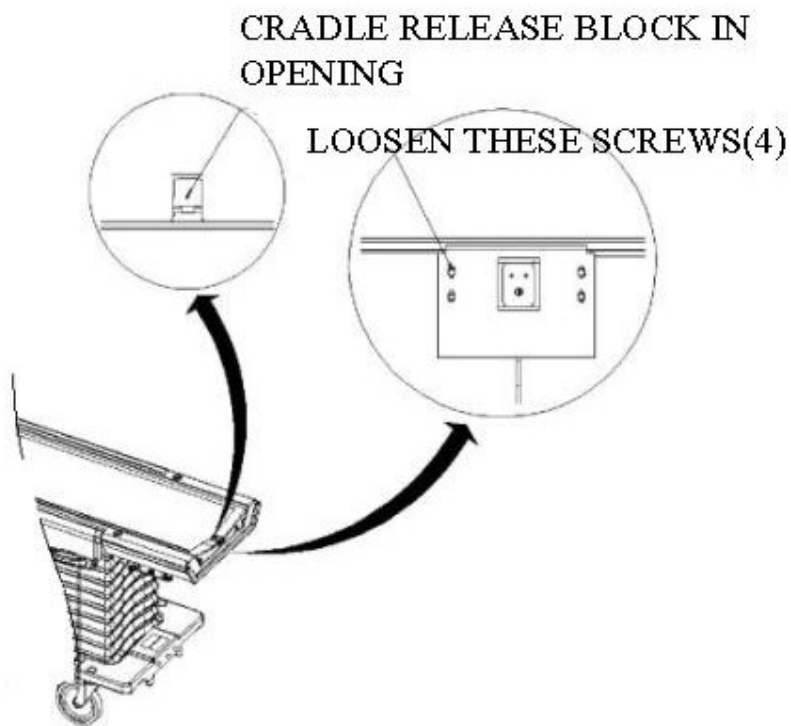


ILLUSTRATION 59

**17.4 Disassembling procedure of the latch assembly holding blocks**

Loosen four hex socket head screws securing the release block assembly.

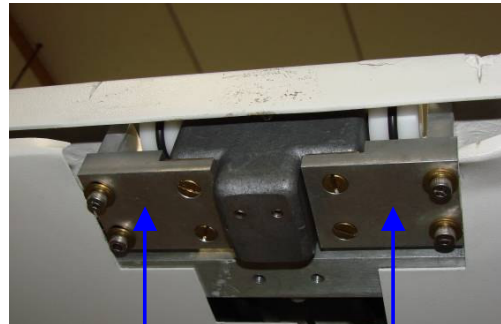
See Illustration 60.



**CRADLE RELEASE BLOCK POSITION ADJUSTMENT  
ILLUSTRATION 60**



**Front latch assembly with guide rollers**



Holding block-1

Holding block-2

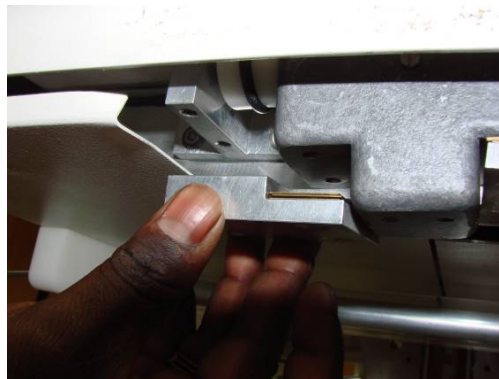
### ILLUSTRATION 61

Assemble the holding blocks to tabletop using disassembled Allen screws as shown in ILLUSTRATION 61



### ILLUSTRATION 62

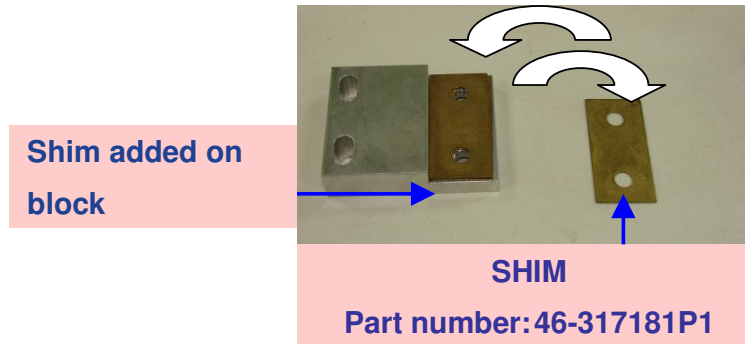
Loosen the screw from holding block-1 using screwdriver as shown in ILLUSTRATION 62.



### ILLUSTRATION 63

Remove the block1 as shown in ILLUSTRATION 63,same procedure to be followed to remove the other block.

**17.5 Additions or Deletion of Shims:**



**ILLUSTRATION 64**

Add or remove shims on the blocks to increase or decrease the height, until the o-rings are .25"(6.35mm) Above horizontal wheel track on either side as shown in ILLUSTRATION 64.

**17.6 Assembling procedure of the latch assembly holding blocks**



**ILLUSTRATION 65.**

Tighten the screw of the holding blocks using screwdriver as shown in ILLUSTRATION 65 and check for the Height of rollers by assembling holding blocks to Table top as shown in ILLUSTRATION 60, if height is ok go to next step, if height is not ok follow the steps 17.4 to 17.6

**17.7 Put back the cover using Four Allen screws as shown in ILLUSTRATION 60.**

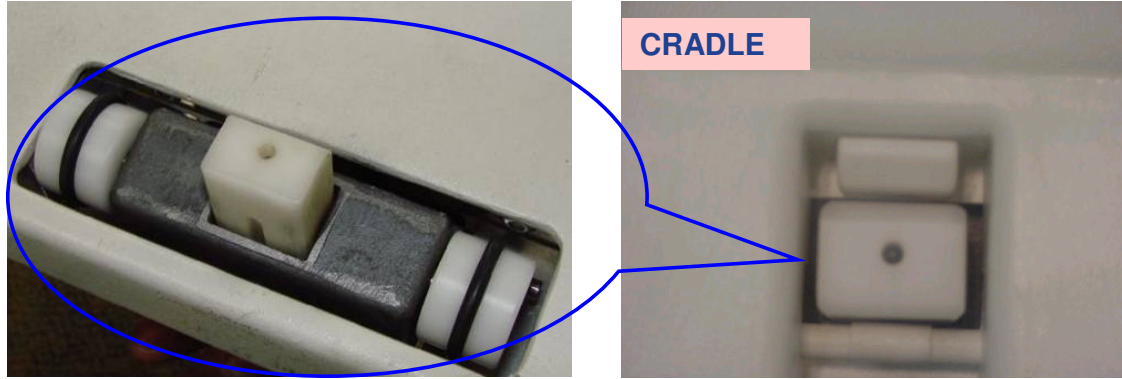


ILLUSTRATION 66



**CAUTION!** Ensure that the front Latch is centralized in the Slot of the cradle as shown in ILLUSTRATION 66

**NOTE:** The front latch height has to be readjusted after setting the height of the roller, Check the delrin block is flushed or 2mm projection with the casting in the Dock condition, if not follow Section 7.2 Cradle release block Retraction adjustment.

## ANNEXURE 1

The below Procedure holds good for Signa Lite tables with the following Table Serial Nos. only **2815WH9, 2816WH7, 2817WH5, 2818WH3, 2819WH1, 2820WH9, 2821WH7.**

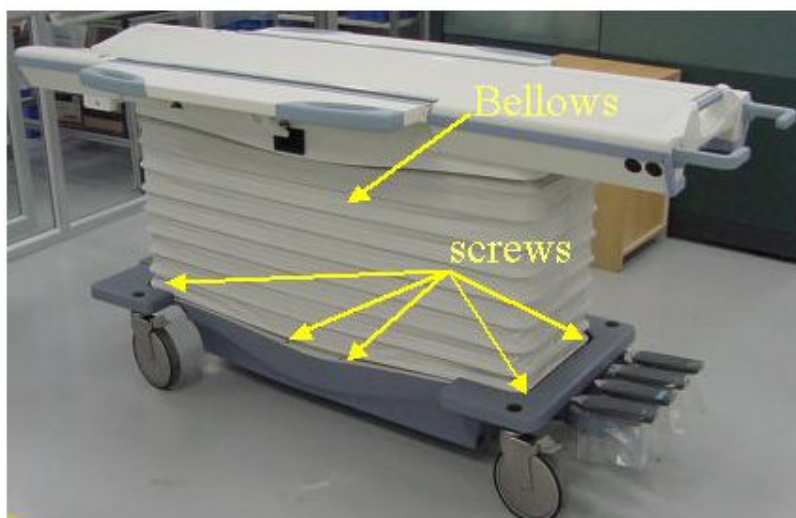
### 1 PULLING UP OF BELLOWS TO SERVICE POSITION

See ILLUSTRATION A for location of Bellow on patient table and its Attached hardware.

- 1.1 Undock patient table, and move it out of exam room.
- 1.2 Remove the 8 screws on either side of the table as shown in illustration A (screws on the other side is not shown)

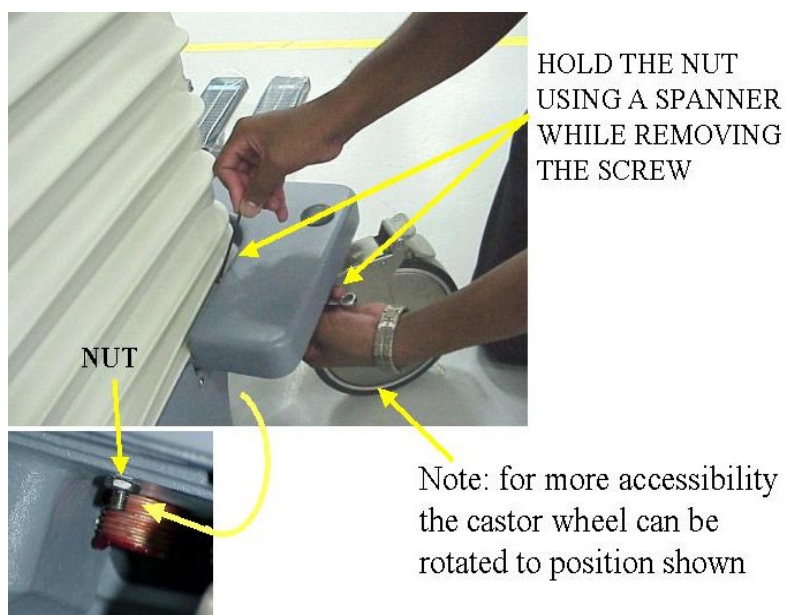
**Note: The 4 corner screws fixing the bellows on the casting is fixed using a nut, to remove these insert an Allen key into the screw head and rotate the nut using a spanner. Refer to illustration B.**

- 1.3 Remove all screws and disengage the bottom rim of the bellows from the Velcro.



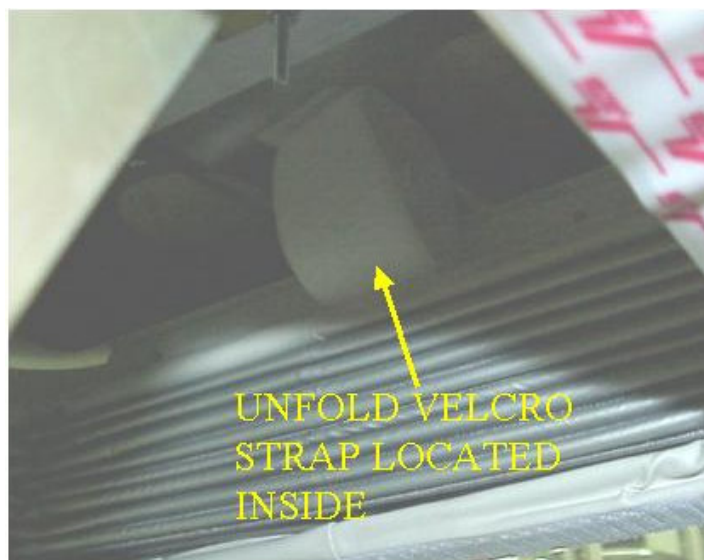
**BELLOWS SCREW REMOVAL  
ILLUSTRATION A**





**ILLUSTRATION B**

- 1.4 Lift Bellows upwards and Strap it using Velcro straps provided at the top. (The straps are located inside bellows. Refer to illustration C). Refer illustration D and E.



**ILLUSTRATION C**

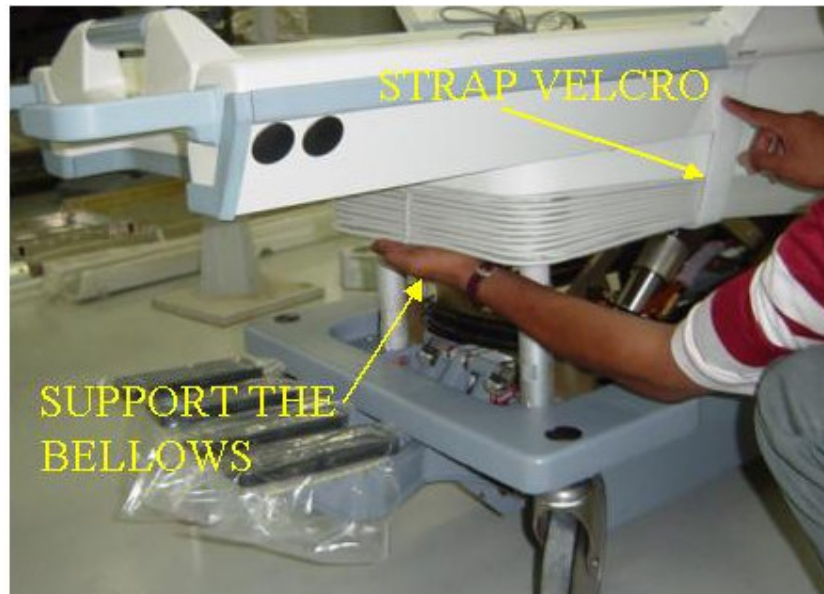


ILLUSTRATION D



**FASTENING OF BELLOWS TO SERVICE POSITION**

ILLUSTRATION E



**CAUTION!**

**DO NOT MOVE THE TABLE UP/DOWN WHEN THE BELLOWS IS IN SERVICE POSITION**

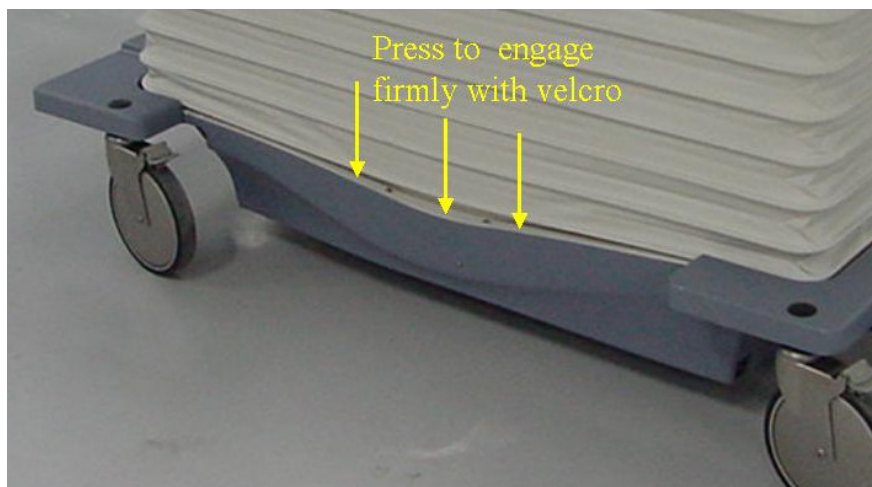
## 2 MOVING BELLOWS TO WORKING/NORMAL POSITION.

- 2.1 Unfasten all the Velcro straps from the top condition and bring the bellows to Bottom position.
- 2.2 Here make sure to fold back the Velcro straps and fasten inside as shown in Illustration F, otherwise the Velcro strap can get entangled while the bellows is in up/down motion.



**ILLUSTRATION F**

- 2.3 Position the bottom rim of the bellows on to the casting and fasten the screws at the four corners firmly. Refer illustration A and B. Also fasten screws connecting the bellows to the bottom cover. Refer Illustration G.



**FASTENING OF BELLOWS TO BOTTOM POSITION**

**ILLUSTRATION G**

**ANNEXURE 2.**

Note: Below process is for table serial Nos 2816WH7, 2817WH5, 2818WH3, 2819WH1 which have old design of reservoir.

**1 HYDRAULIC FLUID LEVEL CHECK AND FILL.**

- 1.1 Move bellows to service/up position. (Refer annexure 1).
- 1.2 Raise table to FULL UP position.
- 1.3 Check hydraulic fluid level in reservoir at the oil level mark. Add fluid if necessary by removing the small cap.
- 1.4 Move the bellows back to normal working condition. Refer to annexure 1



**HYDRAULIC FLUID LEVEL CHECK  
ILLUSTRATION H**